

# Continuum-Mechanics-MFEM

## Theory Manual of the Flux-Kernel Framework (`src/`)

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### Abstract

This manual describes the mathematical models and the numerical methods implemented in the root `src/` library of the repository: finite-strain solid mechanics in total Lagrangian form on a fixed reference mesh, quasi-static or, when the input asks for it, with inertia, discretised with continuous Galerkin finite elements on MFEM. It covers the kinematics and balance laws, the hyperelastic constitutive library in four categories (compressible coupled, compressible uncoupled, slightly compressible and incompressible), small-strain linear elasticity as a special case of the same forms, the two-dimensional plane-strain and plane-stress reductions, the displacement and the mixed displacement–pressure formulations, boundary conditions and load scheduling, the inertial term and its implicit time integration, the element kernels and the dual-number tangents, the load and time stepper, damped Newton, the algebraic multigrid and saddle-point linear solvers, and the post-processing of quadrature-point quantities. The verification and validation of the implementation against closed-form and reference solutions is documented separately in the verification manual (`doc/verification_manual.tex`), which covers the inputs and scripts under `apps/`. The legacy applications under `myapps/` are not part of this document.

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# 1 Introduction

## 1.1 Scope and architecture

The framework is a multiphysics library on MFEM in which each physics is meant to be expressed as fluxes and sources evaluated at quadrature points, while the framework owns assembly, the nonlinear solver and the linear solvers (the design is laid out in `doc/flux_kernel_architecture.html` and the implementation plan in `doc/hyperelasticity_implementation_plan.md`). The first, and at present only, physics is quasi-static nonlinear solid mechanics. The library is organised in five layers:

| Layer                               | Content                                                                                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>src/base/</code>              | fixed-size tensors, forward-mode dual numbers, the YAML schema and its validation, the expression language for boundary data, mesh input, a named field registry, ParaView output and point probes |
| <code>src/kernels/</code>           | the total Lagrangian element integrators (displacement and mixed $u$ - $p$ ) and the follower-pressure boundary face integrator                                                                    |
| <code>src/kernels/materials/</code> | the hyperelastic material functors and the small-strain linear elastic one, the kinematics trait, the plane-stress adapter, the dual-number tangent, the material variant and the YAML factory     |
| <code>src/physics/</code>           | the solid mechanics modules (residual, Jacobian, fields), the load set shared by both formulations, and the quadrature-point post-processing                                                       |
| <code>src/solvers/</code>           | the quasi-static load stepper, damped Newton, the Krylov-AMG linear solver and the augmented-Lagrangian saddle-point solver                                                                        |

Executables under `apps/` only parse the input and wire library objects together; they contain no physics, assembly or solver code. The executable `apps/solid_mechanics.cpp` and its inputs are described in the verification manual.

## 1.2 What is implemented

- Balance of linear momentum in total Lagrangian form: the mesh never moves, every integral is over the reference configuration. By default the analysis is quasi-static: loads and prescribed displacements are functions of a pseudo-time  $t \in [0, 1]$  and inertia is absent.
- Elastodynamics on request (Section 8): the inertial term  $\int_{\Omega_R} \rho_R \ddot{\mathbf{u}} \cdot \mathbf{w} \, dV$  with a constant, consistent mass matrix and the reference density by element attribute, integrated in physical time by the generalized- $\alpha$  family in displacement form (Newmark, HHT- $\alpha$ , generalized- $\alpha$ ), for every material and both formulations. It is a decorator over the quasi-static problem, not a second set of kernels.
- Hyperelasticity in four categories (Section 4.1): compressible coupled models (neo-Hookean of Bonet–Wood form, Saint Venant–Kirchhoff); compressible uncoupled models  $\Psi_{\text{iso}}(\bar{\mathbf{F}}) + \kappa u(J)$  with a selectable volumetric law  $u(J)$ , quadratic  $\frac{1}{2}(J-1)^2$  by default (isochoric neo-Hookean, Mooney–Rivlin, Yeoh, Gent, Arruda–Boyce, Ogden); the same six as slightly compressible materials with the pressure as an independent field; and the same six as incompressible materials with  $\kappa = \infty$  and a Lagrange multiplier. Materials are stateless and may vary by element attribute.

- Small-strain (geometrically linear) isotropic elasticity, `linear_elastic`, in both formulations, including  $\nu = \frac{1}{2}$  (Section 7). It is a material of the same kernels, with the kinematics as a compile-time trait; the problem is linear and is solved by one Newton step.
- Two formulations: a pure displacement formulation with the assembled tangent, and a mixed displacement–pressure formulation on Taylor–Hood pairs for near- and fully incompressible materials.
- Two-dimensional problems as plane strain (default) or plane stress (displacement formulation; the thickness stretch is eliminated pointwise at the material level).
- Boundary conditions: prescribed displacements on all or some components, nominal tractions, dead normal pressures, follower pressures per unit current area, and a body force per unit mass, each with its own load schedule and data given as expressions of the reference coordinates and the pseudo-time.
- Consistent tangents from forward-mode automatic differentiation of the same code that evaluates the stress (Ogden and the compressible plane-stress adapter use analytic or implicit derivatives where eigenvectors or a root are not differentiable by duals alone).
- Solvers: load stepping, and time stepping, with bisection on failure, damped Newton with Armijo backtracking, GMRES or CG with BoomerAMG, and FGMRES with a block upper-triangular preconditioner for the  $u$ – $p$  saddle point, whose Schur complement approximation gains an inertial part in a dynamic analysis.
- Post-processing of Cauchy and first Piola–Kirchhoff stress, the deformation gradient, the strain, the volume ratio, von Mises stress, energy density and thickness stretch at quadrature points, presented as nodal fields, element averages or point clouds; probes; reactions of the Dirichlet entries; in a dynamic analysis the nodal velocity and acceleration, and the kinetic and internal energy with the external work.

Not implemented: explicit time integration, physical damping, modal analysis, internal variables and inelasticity, temperature dependence, contact, point loads, multi-point or periodic constraints, discontinuous Galerkin or hybridised discretisations, partial assembly, and device execution. Section 15 lists what of the present code is specific to the solid and what a second physics would generalise.

### 1.3 Conventions and companion notes

Referential (Lagrangian) quantities carry the subscript  $R$ ; the current configuration is  $\Omega_t$ . The left Cauchy–Green tensor is  $\mathbf{B} = \mathbf{F}\mathbf{F}^T$ , the right one  $\mathbf{C} = \mathbf{F}^T\mathbf{F}$ . The pressure unknown of the mixed formulation and the `pressure` output field are the tensile-positive mean stress  $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$ . In two dimensions every  $3 \times 3$  tensor is built from the in-plane block with  $F_{33} = 1$  (plane strain) or  $F_{33} = \lambda_3$  (plane stress).

The verification manual (`doc/verification_manual.tex`) documents every input and test; its appendix collects the closed-form homogeneous states of the incompressible models that serve as reference solutions.

## 2 Kinematics

Let  $\Omega_R \subset \mathbb{R}^d$ ,  $d \in \{2, 3\}$ , be the reference configuration with boundary  $\partial\Omega_R$  and outward unit normal  $\mathbf{N}$ . The motion  $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t)$  carries the material point  $\mathbf{X}$  to  $\mathbf{x}$ ;  $t$  is a pseudo-time that parametrises the load path (Section 9). The deformation gradient, its determinant and the

strain measures are

$$\mathbf{F} = \text{Grad } \boldsymbol{\varphi} = \mathbf{I} + \mathbf{H}, \quad \mathbf{H} = \text{Grad } \mathbf{u}, \quad J = \det \mathbf{F} > 0, \quad (1)$$

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{B} = \mathbf{F} \mathbf{F}^T, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad (2)$$

with  $\text{Grad} = \partial/\partial \mathbf{X}$ . The principal invariants of  $\mathbf{C}$  (equivalently of  $\mathbf{B}$ ) are

$$I_1 = \text{tr } \mathbf{C}, \quad I_2 = \frac{1}{2}[(\text{tr } \mathbf{C})^2 - \text{tr}(\mathbf{C}^2)], \quad I_3 = \det \mathbf{C} = J^2, \quad (3)$$

and in terms of the principal stretches  $\lambda_a$  (square roots of the eigenvalues  $c_a$  of  $\mathbf{C}$ )  $I_1 = \sum_a \lambda_a^2$ ,  $I_2 = \sum_{a < b} \lambda_a^2 \lambda_b^2$ ,  $J = \lambda_1 \lambda_2 \lambda_3$ . The isochoric (volume-preserving) part of the deformation and the modified invariants are

$$\bar{\mathbf{F}} = J^{-1/3} \mathbf{F}, \quad \bar{I}_1 = J^{-2/3} I_1, \quad \bar{I}_2 = J^{-4/3} I_2, \quad \bar{\lambda}_a = J^{-1/3} \lambda_a. \quad (4)$$

The stress measures are the Cauchy stress  $\boldsymbol{\sigma}$  (force per unit current area), the first Piola–Kirchhoff (nominal) stress  $\mathbf{P}$  (force per unit reference area) and the second Piola–Kirchhoff stress  $\mathbf{S}$ , related by

$$\mathbf{P} = J \boldsymbol{\sigma} \mathbf{F}^{-T}, \quad \mathbf{S} = \mathbf{F}^{-1} \mathbf{P}, \quad \boldsymbol{\sigma} = J^{-1} \mathbf{P} \mathbf{F}^T. \quad (5)$$

$\boldsymbol{\sigma}$  and  $\mathbf{S}$  are symmetric;  $\mathbf{P}$  is not, but satisfies  $\mathbf{P} \mathbf{F}^T = \mathbf{F} \mathbf{P}^T$ . Nanson’s formula relates area elements,

$$\mathbf{n} \, da = J \mathbf{F}^{-T} \mathbf{N} \, dA = \text{cof } \mathbf{F} \mathbf{N} \, dA, \quad (6)$$

and is what turns a traction per unit current area into a nominal traction. The reference density is  $\rho_R$ ; mass conservation gives  $\rho_R = J \rho$ .

In two dimensions the kernels form the  $2 \times 2$  displacement gradient  $\mathbf{H}_{2D}$  and embed it as

$$\mathbf{F} = \begin{pmatrix} \mathbf{I}_2 + \mathbf{H}_{2D} & \mathbf{0} \\ \mathbf{0}^T & F_{33} \end{pmatrix}, \quad F_{33} = 1 \text{ (plane strain)}, \quad F_{33} = \lambda_3 \text{ (plane stress, Section 5)}, \quad (7)$$

so that one three-dimensional constitutive code path serves every dimension.

### 3 Balance of momentum and the displacement weak form

#### 3.1 Strong form

Quasi-static balance of linear momentum on the reference configuration reads

$$\text{Div } \mathbf{P} + \rho_R \mathbf{b} = \mathbf{0} \quad \text{in } \Omega_R, \quad (8a)$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_R^D, \quad (8b)$$

$$\mathbf{P} \mathbf{N} = \bar{\mathbf{T}} \quad \text{on } \Gamma_R^N, \quad (8c)$$

where  $\mathbf{b}$  is the body force per unit mass,  $\bar{\mathbf{u}}$  the prescribed displacement (possibly of a subset of components) and  $\bar{\mathbf{T}}$  the prescribed nominal traction. Balance of angular momentum is  $\mathbf{P} \mathbf{F}^T = \mathbf{F} \mathbf{P}^T$  and is satisfied identically by hyperelastic stresses. The system is closed by a hyperelastic constitutive law

$$\mathbf{P} = \frac{\partial \Psi}{\partial \mathbf{F}}(\mathbf{F}), \quad (9)$$

with  $\Psi$  the stored energy per unit reference volume (Section 4).

### 3.2 Weak form

With the trial and test spaces on the fixed reference domain,

$$\mathcal{S} = \{\mathbf{u} \in [H^1(\Omega_R)]^d : \mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_R^D\}, \quad \mathcal{V} = \{\mathbf{w} \in [H^1(\Omega_R)]^d : \mathbf{w} = \mathbf{0} \text{ on } \Gamma_R^D\}, \quad (10)$$

the total Lagrangian weak form is: find  $\mathbf{u} \in \mathcal{S}$  such that for all  $\mathbf{w} \in \mathcal{V}$

$$\boxed{R(\mathbf{u}; \mathbf{w}) = \int_{\Omega_R} \mathbf{P}(\mathbf{F}) : \text{Grad } \mathbf{w} \, dV - \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV - \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA + R_{\text{fol}}(\mathbf{u}; \mathbf{w}) = 0,} \quad (11)$$

where  $R_{\text{fol}}$  is the contribution of follower pressures (Section 9.4), zero when there are none. The sign convention throughout the code is *residual = internal minus external*. The form is nonlinear in  $\mathbf{u}$  through  $\mathbf{P}(\mathbf{F})$  and through the geometry in  $\mathbf{F}$ .

**Equivalent forms.** With  $\mathbf{P} = \mathbf{F}\mathbf{S}$  and the symmetry of the second Piola–Kirchhoff stress,  $\mathbf{P} : \text{Grad } \mathbf{w} = \mathbf{S} : \delta \mathbf{E}[\mathbf{w}]$ , where  $\delta \mathbf{E}[\mathbf{w}] = \frac{1}{2}(\mathbf{F}^T \text{Grad } \mathbf{w} + \text{Grad}^T \mathbf{w} \mathbf{F})$  is the variation of the Green–Lagrange strain, so the internal virtual work may be written  $\int_{\Omega_R} \mathbf{S}(\mathbf{E}) : \delta \mathbf{E}[\mathbf{w}] \, dV$ , the form commonly used with  $(\mathbf{S}, \mathbf{E})$  constitutive pairs such as Saint Venant–Kirchhoff; its linearisation splits into the familiar material and geometric stiffness, which the nominal tangent  $\mathbb{A}$  of (13) contains together. Pushing forward with  $\mathbf{P} : \text{Grad } \mathbf{w} = J\boldsymbol{\sigma} : \text{grad } \mathbf{w}$  and  $dv = J \, dV$  gives the spatial statement on the current configuration  $\Omega_t$ ,

$$\int_{\Omega_t} \boldsymbol{\sigma} : \text{grad } \mathbf{w} \, dv = \int_{\Omega_t} \rho \mathbf{b} \cdot \mathbf{w} \, dv + \int_{\Gamma_t^N} \bar{\mathbf{t}} \cdot \mathbf{w} \, da, \quad \text{div } \boldsymbol{\sigma} + \rho \mathbf{b} = \mathbf{0} \text{ in } \Omega_t, \quad \boldsymbol{\sigma} \mathbf{n} = \bar{\mathbf{t}} \text{ on } \Gamma_t^N, \quad \boldsymbol{\sigma} = \boldsymbol{\sigma}^T, \quad (12)$$

with  $\rho = \rho_R/J$  and  $\bar{\mathbf{t}}$  the traction per unit current area: the two statements are identical and differ only in the configuration on which the integrals are evaluated. The code integrates on  $\Omega_R$  throughout. With inertia the residual gains  $\int_{\Omega_R} \rho_R \ddot{\mathbf{u}} \cdot \mathbf{w} \, dV$  (spatially  $\rho \dot{\mathbf{v}}$  with the material acceleration  $\dot{\mathbf{v}} = \partial_t \mathbf{v} + (\text{grad } \mathbf{v})\mathbf{v}$ ) and needs initial data  $\mathbf{u}_0, \mathbf{v}_0$ : Section 8.

### 3.3 Linearisation

Newton’s method needs the directional derivative of (11). For the internal term,

$$D_{\Delta \mathbf{u}} \int_{\Omega_R} \mathbf{P} : \text{Grad } \mathbf{w} \, dV = \int_{\Omega_R} \text{Grad } \mathbf{w} : \mathbb{A} : \text{Grad } \Delta \mathbf{u} \, dV, \quad \mathbb{A} = \frac{\partial \mathbf{P}}{\partial \mathbf{F}} = \frac{\partial^2 \Psi}{\partial \mathbf{F} \partial \mathbf{F}}, \quad A_{ijkl} = \frac{\partial P_{ij}}{\partial F_{kl}}. \quad (13)$$

$\mathbb{A}$  is the fourth-order material tangent in nominal form. For a hyperelastic material it has the major symmetry  $A_{ijkl} = A_{klij}$ , so the assembled stiffness is symmetric; the follower pressure adds a non-symmetric part (Section 9.4). The code never writes  $\mathbb{A}$  by hand: it is produced by differentiating the stress evaluation with dual numbers (Section 11).

## 4 Constitutive models

Every material in the library is a stateless value type whose parameters are public members and whose stress function is a template over the scalar type,

$$\mathbf{P} = \text{PK1}(\mathbf{F}) : \text{tensor}\langle T, 3, 3 \rangle \rightarrow \text{tensor}\langle T, 3, 3 \rangle, \quad T \in \{\text{double}, \text{dual}\}, \quad (14)$$

optionally with an energy  $\Psi(\mathbf{F})$ . Models for the mixed formulation expose the split  $\mathbf{P}_{\text{iso}}$ ,  $\Psi_{\text{iso}}$ , the volumetric pressure  $U'(J)$ , the bulk modulus  $\kappa$  (possibly infinite) and a small-strain shear modulus used to scale the saddle-point preconditioner.

#### 4.1 Classification by the treatment of volume change

All models are isotropic and hyperelastic; what distinguishes them is how the volume change  $J$  enters the energy and how the resulting pressure is handled numerically. Four categories are used in this manual:

| Category                         | Volumetric treatment                                                                                                                                     | Models                                                                                                                                             | Formulation and input                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1.<br>Compressible,<br>coupled   | one energy $\Psi(\mathbf{F})$ ; $J$ enters through $\ln J$ or $\text{tr } \mathbf{E}$ terms mixed with the shear response                                | <code>neo_hookean</code> , <code>st_venant_kirchhoff</code> , <code>gent_compressible_summit</code>                                                | displacement; <code>E</code> , <code>nu</code> with $\nu < \frac{1}{2}$                                  |
| 2.<br>Compressible,<br>uncoupled | $\Psi_{\text{iso}}(\bar{\mathbf{F}}) + \kappa u(J)$ , $u$ one of four laws (quadratic $\frac{1}{2}(J-1)^2$ by default); $\kappa/\mu$ of order one to ten | <code>iso_neo_hookean</code> , <code>mooney_rivlin</code> , <code>yeoh</code> , <code>gent</code> , <code>arruda_boyce</code> , <code>ogden</code> | displacement (mixed also allowed); <code>kappa</code> or <code>nu</code>                                 |
| 3. Slightly<br>compressible      | the same split with $\kappa \gg \mu$ ; the pressure $p = U'(J)$ becomes an independent field                                                             | the same six                                                                                                                                       | mixed (the displacement formulation locks); <code>kappa</code> or <code>nu</code> close to $\frac{1}{2}$ |
| 4.<br>Incompressible             | $J = 1$ enforced by a Lagrange multiplier $p$ ; $\kappa = \infty$                                                                                        | the same six                                                                                                                                       | mixed, or plane stress in 2D; <code>incompressible: true</code> or <code>nu: 0.5</code>                  |

Categories 2, 3 and 4 share one code path: the six decoupled materials provide  $\Psi_{\text{iso}}$  and  $U'(J)$ , and the category is set by the value of  $\kappa$  and by the formulation chosen in the input. The border between 2 and 3 is one of degree, not of kind: the displacement formulation is adequate up to  $\kappa/\mu$  of order ten ( $\nu \approx 0.45$ , where the tests confirm that both formulations agree under refinement), while at  $\nu = 0.4999$  ( $\kappa/\mu \approx 5000$ ) the mixed formulation is required in practice. Category 1 is a different code path with no split; those two models cannot be made incompressible.

#### 4.2 Category 1: compressible, coupled

**Neo-Hookean (Bonet–Wood form).**

$$\Psi = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda}{2}(\ln J)^2, \quad \mathbf{P} = \mu(\mathbf{F} - \mathbf{F}^{-T}) + \lambda \ln J \mathbf{F}^{-T}. \quad (15)$$

This is the compressible extension of the neo-Hookean energy given by Bonet and Wood [1] and used by the FEniCS and FEniCSx hyperelasticity demos and by FEBio (the source comment also names Simo–Hughes; the exact expression is not confirmed there). The  $-\mu \ln J$  term cancels the stress of the  $\mu \mathbf{F}$  term at  $\mathbf{F} = \mathbf{I}$ , so the reference state is stress free; the small-strain limit is linear elasticity with the Lamé constants  $(\mu, \lambda)$ ; and both logarithmic terms diverge as  $J \rightarrow 0$ , so compression to zero volume costs infinite energy. The shear term uses the full  $I_1$ , not  $\bar{I}_1$ , so shear and volume are coupled and the volumetric response is  $\lambda \ln J$  rather than a separate law  $U(J)$ . Other coupled compressible neo-Hookean forms in the literature replace  $\frac{\lambda}{2}(\ln J)^2$  by  $\frac{\lambda}{2}(J-1)^2$  or by the polyconvex Ciarlet–Geymonat combination  $\frac{\Lambda}{4}(J^2 - 1) - \frac{\Lambda}{2} \ln J$  [5]; none of these is implemented.

### Saint Venant–Kirchhoff.

$$\Psi = \frac{\lambda}{2}(\text{tr } \mathbf{E})^2 + \mu \mathbf{E} : \mathbf{E}, \quad \mathbf{S} = \lambda \text{tr } \mathbf{E} \mathbf{I} + 2\mu \mathbf{E}, \quad \mathbf{P} = \mathbf{F} \mathbf{S}. \quad (16)$$

This is linear elasticity in the Green–Lagrange strain; it is not polyconvex, its energy stays finite as  $J \rightarrow 0$  and it softens in compression, so it is used chiefly for manufactured-solution tests.

### Compressible Gent, SUMMIT form (`gent_compressible_summit`).

$$\Psi = -\frac{\mu}{2} \left( J_m \ln \left( 1 - \frac{I_1 - 3}{J_m} \right) + 2 \ln J \right) + \frac{\kappa}{2} A^4, \quad A = \frac{J^2 - 1}{2} - \ln J, \quad (17)$$

$$\mathbf{P} = \frac{\mu J_m}{J_m - I_1 + 3} \mathbf{F} + (2\kappa A^3(J^2 - 1) - \mu) \mathbf{F}^{-T}, \quad I_1 - 3 < J_m. \quad (18)$$

This is the `gent-compressible` material of the SUMMIT code (`summit::GentCompressibleHyperelastic`), transcribed so that results can be compared between the two codes: the Gent limited-extensibility term on the full  $I_1$  (not  $\bar{I}_1$ , unlike the decoupled `gent` of Section 4.3), the  $-\mu \ln J$  term that makes the reference state stress free, and a quartic volumetric penalty. For  $J_m \rightarrow \infty$  the chain term reduces to the neo-Hookean  $\frac{\mu}{2}(I_1 - 3)$ . Since  $A = (J - 1)^2 + O((J - 1)^3)$ , the penalty is  $O((J - 1)^8)$  and  $\kappa$  does *not* enter the small-strain response: the linearised moduli are  $\mu$  and  $\lambda = 2\mu/J_m$  (the latter from the chain term, because  $I_1 - 3 \approx 2 \text{tr } \boldsymbol{\varepsilon}$ ), that is  $\nu = 1/(J_m + 2)$ , whatever  $\kappa$  is.  $\kappa$  is therefore a penalty parameter that stiffens the volume change at finite strain only, not a bulk modulus; SUMMIT reports  $E$  and  $\nu$  computed from  $\kappa$  and  $\mu$  as if it were one (they enter only its wave-speed estimate). The input keys are `mu`, `kappa` and `Jm`; the tangent comes from automatic differentiation and agrees with SUMMIT’s closed-form moduli to round-off (`test_materials`). Displacement formulation only.

The first two take  $(E, \nu)$  from the input and convert with  $\mu = E/(2(1 + \nu))$ ,  $\lambda = E\nu/((1 + \nu)(1 - 2\nu))$ ; the derived bulk modulus  $\kappa = \lambda + \frac{2}{3}\mu$  is reported but does not enter the stress. None of the three models separates the pressure, so none can be used in the mixed formulation or in the incompressible limit.

### 4.3 Category 2: compressible, uncoupled

The remaining six models are decoupled,

$$\Psi(\mathbf{F}) = \Psi_{\text{iso}}(\bar{\mathbf{F}}) + U(J), \quad U(J) = \kappa u(J), \quad u(J) = \frac{1}{2}(J - 1)^2 \text{ by default}, \quad (19)$$

where  $\Psi_{\text{iso}}$  depends on  $\mathbf{F}$  only through the isochoric part (4) and  $u(J)$  is one of the volumetric laws of the paragraph below, normalised so that  $u''(1) = 1$ . Because  $\Psi_{\text{iso}}$  is invariant under  $\mathbf{F} \rightarrow \alpha \mathbf{F}$ , Euler’s theorem gives  $\mathbf{P}_{\text{iso}} : \mathbf{F} = 0$ , i.e. the isochoric Cauchy stress  $J^{-1} \mathbf{P}_{\text{iso}} \mathbf{F}^T$  is deviatoric for every  $\mathbf{F}$ , not only at  $J = 1$ . In the displacement formulation the stress is

$$\mathbf{P} = \mathbf{P}_{\text{iso}}(\mathbf{F}) + U'(J) J \mathbf{F}^{-T}, \quad \boldsymbol{\sigma} = J^{-1} \mathbf{P}_{\text{iso}} \mathbf{F}^T + U'(J) \mathbf{I}, \quad U'(J) = \kappa u'(J) = \kappa(J - 1) \text{ by default}, \quad (20)$$

so  $p = U'(J)$  is exactly the mean stress  $\frac{1}{3} \text{tr } \boldsymbol{\sigma}$ . At moderate  $\kappa/\mu$  this is an ordinary compressible material and the pure displacement discretisation is well behaved.

**Volumetric law.** The volumetric law is selectable per material (`material.volumetric`, `materials/volumetric.hpp`). Writing  $U(J) = \kappa u(J)$ , the four laws are

| Key                      | $u(J)$                           | $u'(J)$                   | $u''(J)$                  | Notes                                                 |
|--------------------------|----------------------------------|---------------------------|---------------------------|-------------------------------------------------------|
| <code>quadratic</code>   | $\frac{1}{2}(J-1)^2$             | $J-1$                     | 1                         | default; $U$ finite as $J \rightarrow 0$              |
| <code>simo_taylor</code> | $\frac{1}{4}(J^2 - 1 - 2 \ln J)$ | $\frac{1}{2}(J - J^{-1})$ | $\frac{1}{2}(1 + J^{-2})$ | Simo–Taylor; convex, $U \rightarrow \infty$ both ways |
| <code>logarithmic</code> | $\frac{1}{2}(\ln J)^2$           | $\ln J/J$                 | $(1 - \ln J)/J^2$         | Anand’s FEniCSx codes; $u'$ peaks at $J = e$          |
| <code>j_log_j</code>     | $J \ln J - J + 1$                | $\ln J$                   | $1/J$                     | polyconvex; $u'$ monotone                             |

The logarithmic law softens in dilatation: its tangent bulk modulus  $\kappa(1 - \ln J)/J^2$  falls to  $0.18\kappa$  at  $J = 1.67$  and to zero at  $J = e$ , where the pressure peaks at  $\kappa/e$ . Its Newton iterates must therefore stay near  $J = 1$  (run it with the tangent predictor of Section 12), and it can fail at stress singularities where  $J$  grows locally, whatever the increment size; the monotone laws do not have this limitation. Every law satisfies  $u(1) = u'(1) = 0$  and  $u''(1) = 1$ , so  $\kappa$  is the small-strain bulk modulus for all of them and the input semantics of `kappa` and `nu` do not change; the laws agree to first order in the strain and differ at finite strain, chiefly in strong compression and in the pressure of nearly incompressible problems (Doll and Schweizerhof [6] survey the family). The displacement formulation needs only  $U'(J) = \kappa u'(J)$ ; the mixed formulation uses the normalised derivatives  $u'$  and  $u''$  (Sections 4.4 and 6). Regions inherit the law of the base material. The law is irrelevant at  $\kappa = \infty$  (category 4) and is not a key of the coupled models.

**Isochoric energies.** The six isochoric energies below are shared by categories 2, 3 and 4.

#### 4.3.1 Models depending on $\bar{I}_1$ alone

For  $\Psi_{\text{iso}} = f(\bar{I}_1)$ ,

$$\mathbf{P}_{\text{iso}} = f'(\bar{I}_1) \frac{\partial \bar{I}_1}{\partial \mathbf{F}}, \quad \frac{\partial \bar{I}_1}{\partial \mathbf{F}} = J^{-2/3} \left( 2\mathbf{F} - \frac{2}{3} I_1 \mathbf{F}^{-T} \right), \quad (21)$$

so that a model only has to provide  $f'$  (`isochoric.hpp`). The library contains

| Model                 | $\Psi_{\text{iso}}$                                        | $f'(\bar{I}_1) = \partial \Psi_{\text{iso}} / \partial \bar{I}_1$ | small-strain $\mu$ |
|-----------------------|------------------------------------------------------------|-------------------------------------------------------------------|--------------------|
| Isochoric neo-Hookean | $\frac{\mu}{2}(\bar{I}_1 - 3)$                             | $\frac{\mu}{2}$                                                   | $\mu$              |
| Yeoh                  | $\sum_{i=1}^3 c_{i0}(\bar{I}_1 - 3)^i$                     | $c_{10} + 2c_{20}(\bar{I}_1 - 3) + 3c_{30}(\bar{I}_1 - 3)^2$      | $2c_{10}$          |
| Gent                  | $-\frac{\mu J_m}{2} \ln(1 - \frac{\bar{I}_1 - 3}{J_m})$    | $\frac{\mu}{2} \frac{J_m}{J_m - (\bar{I}_1 - 3)}$                 | $\mu$              |
| Arruda–Boyce          | $\mu \sum_{i=1}^5 \frac{C_i}{N^{i-1}} (\bar{I}_1^i - 3^i)$ | $\mu \sum_{i=1}^5 \frac{i C_i}{N^{i-1}} \bar{I}_1^{i-1}$          | $\mu_0$ , eq. (22) |

*Isochoric neo-Hookean* (`iso_neo_hookean`): the Gaussian-chain model; the closed form is  $\mathbf{P}_{\text{iso}} = \mu J^{-2/3}(\mathbf{F} - \frac{1}{3} I_1 \mathbf{F}^{-T})$ . *Yeoh* (`yeoh`): a cubic in  $\bar{I}_1 - 3$ ; typically  $c_{20} < 0$ ,  $c_{30} > 0$ , reproducing the softening-then-stiffening response of filled rubbers. *Gent* (`gent`): limited chain extensibility; the energy is defined only for  $\bar{I}_1 - 3 < J_m$ , beyond the locking stretch it evaluates to NaN, which the Newton line search rejects like any non-finite residual, and  $J_m \rightarrow \infty$  recovers the neo-Hookean model. *Arruda–Boyce* (`arruda_boyce`): the eight-chain model in its five-term series in  $\bar{I}_1/N$  with

$C = \{\frac{1}{2}, \frac{1}{20}, \frac{11}{1050}, \frac{19}{7000}, \frac{519}{673750}\}$ ;  $\mu$  is the chain-density parameter  $nkT$ ,  $N$  the number of rigid links per chain (locking stretch  $\lambda_L = \sqrt{N}$ ), and the small-strain shear modulus is

$$\mu_0 = 2f'(3) = \mu \left( 1 + \frac{3}{5N} + \frac{99}{175N^2} + \frac{513}{875N^3} + \frac{42039}{67375N^4} \right). \quad (22)$$

The exact inverse-Langevin form of the model is not implemented; the series is softer near the locking stretch.

#### 4.3.2 Mooney–Rivlin

$$\Psi_{\text{iso}} = c_1(\bar{I}_1 - 3) + c_2(\bar{I}_2 - 3), \quad I_2 = \frac{1}{2}(I_1^2 - \mathbf{C} : \mathbf{C}), \quad (23)$$

$$\mathbf{P}_{\text{iso}} = c_1 J^{-2/3} \left( 2\mathbf{F} - \frac{2}{3} I_1 \mathbf{F}^{-T} \right) + c_2 J^{-4/3} \left( 2I_1 \mathbf{F} - 2\mathbf{F}\mathbf{C} - \frac{4}{3} I_2 \mathbf{F}^{-T} \right), \quad \mu = 2(c_1 + c_2). \quad (24)$$

The only model in the library that depends on the second invariant. With  $c_2 = 0$  it is the isochoric neo-Hookean model.

#### 4.3.3 Ogden

The Ogden energy is a function of the isochoric principal stretches,

$$\Psi_{\text{iso}} = \sum_{r=1}^M \frac{\mu_r}{\alpha_r} (\bar{\lambda}_1^{\alpha_r} + \bar{\lambda}_2^{\alpha_r} + \bar{\lambda}_3^{\alpha_r} - 3), \quad M \leq 6, \quad \mu_r \alpha_r > 0, \quad \mu = \frac{1}{2} \sum_r \mu_r \alpha_r. \quad (25)$$

It is evaluated spectrally. With  $\mathbf{C} = \sum_a c_a \mathbf{N}_a \otimes \mathbf{N}_a$  (cyclic Jacobi eigen-decomposition, `spectral.hpp`) define

$$\beta_a = \sum_r \mu_r \bar{\lambda}_a^{\alpha_r}, \quad \tau_a = \beta_a - \frac{1}{3}(\beta_1 + \beta_2 + \beta_3), \quad \mathbf{S}_{\text{iso}} = \sum_a \frac{\tau_a}{c_a} \mathbf{N}_a \otimes \mathbf{N}_a, \quad \mathbf{P}_{\text{iso}} = \mathbf{F} \mathbf{S}_{\text{iso}}, \quad (26)$$

where  $\tau_a$  are the isochoric principal Kirchhoff stresses (the  $\frac{1}{3} \sum \beta$  term is the deviatoric projection that  $\bar{\lambda}_a = J^{-1/3} \lambda_a$  produces). Eigenvectors are not differentiable at coincident eigenvalues, so the tangent is not obtained by running dual numbers through the eigen-solver. Instead the dual overload of  $\mathbf{P}_{\text{iso}}$  forms the directional derivative along  $d\mathbf{F}$  analytically (Ogden 1984; Simo and Taylor 1991): with  $d\mathbf{C} = d\mathbf{F}^T \mathbf{F} + \mathbf{F}^T d\mathbf{F}$  expressed in the eigenbasis,  $d\hat{\mathbf{C}} = \mathbf{N}^T d\mathbf{C} \mathbf{N}$ , the derivative of  $\mathbf{S}_{\text{iso}}$  in that basis has diagonal entries  $\sum_b (\partial s_a / \partial c_b) d\hat{C}_{bb}$  with  $s_a = \tau_a / c_a$ , and off-diagonal entries

$$d\hat{S}_{ab} = \gamma_{ab} d\hat{C}_{ab}, \quad \gamma_{ab} = \begin{cases} \frac{s_a - s_b}{c_a - c_b}, & |c_a - c_b| > 10^{-8} \max_c c_c, \\ \frac{\partial s_a}{\partial c_a} - \frac{\partial s_a}{\partial c_b}, & \text{otherwise (limit branch)}, \end{cases} \quad (27)$$

followed by  $d\mathbf{S} = \mathbf{N} d\hat{\mathbf{S}} \mathbf{N}^T$  and  $d\mathbf{P} = d\mathbf{F} \mathbf{S}_{\text{iso}} + \mathbf{F} d\mathbf{S}$ . The partial derivatives  $\partial s_a / \partial c_b$  follow from  $\partial \beta_a / \partial c_b = \beta'_a (\delta_{ab} / (2c_a) - 1 / (6c_b))$  with  $\beta'_a = \sum_r \mu_r \alpha_r \bar{\lambda}_a^{\alpha_r}$ . Ogden with one term  $(\mu, 2)$  is the isochoric neo-Hookean model, with two terms  $(2c_1, 2), (-2c_2, -2)$  the Mooney–Rivlin model.

#### 4.4 Category 3: slightly compressible

The same six materials with  $\kappa \gg \mu$  describe rubber-like solids whose bulk modulus is finite but three or four orders of magnitude above the shear modulus (`nu` close to  $\frac{1}{2}$ , or `kappa` given directly). The constitutive law is unchanged, but its numerical treatment is not. In the displacement formulation the volumetric term is a stiff penalty: the discrete field must satisfy  $J \approx 1$  nearly pointwise, which the low-order continuous displacement space cannot do without suppressing admissible deformations (volumetric locking), and the pressure recovered from  $U'(J)$  oscillates because small errors in  $J$  are multiplied by  $\kappa$ . Newton also sees a badly scaled tangent dominated by the penalty.

The remedy is to treat the pressure as an independent field: replace  $U'(J)$  in (20) by an unknown  $p$  and impose the volumetric law weakly,

$$\mathbf{P} = \mathbf{P}_{\text{iso}}(\mathbf{F}) + p J \mathbf{F}^{-T}, \quad \int_{\Omega_R} q \left( u'(J) - \frac{p}{\kappa} \right) dV = 0 \quad \forall q, \quad (28)$$

with  $p$  taken from a lower-order space (Section 6), which the Taylor–Hood pair satisfies without locking. The constraint is written with the normalised pressure  $u' = U'/\kappa$  so that it stays finite at  $\kappa = \infty$ ; for the quadratic law it is the familiar  $J - 1 - p/\kappa$ . For that law the pair is the stationarity of the perturbed Lagrangian  $\int \Psi_{\text{iso}} + p(J - 1) - p^2/(2\kappa) dV$ , the Legendre transform of  $U$ , and the Newton system is symmetric; for the other laws the constraint is not the derivative of a potential in  $(\mathbf{u}, p)$  jointly, and the  $(2, 1)$  block carries the factor  $u''(J)$  (Section 6), which the FGMRES saddle-point solver accepts. Eliminating  $p$  pointwise from (28) recovers (20) exactly for every law, so the mixed and displacement formulations solve the same continuous problem at any finite  $\kappa$ ; their discrete solutions converge to each other under refinement, which the tests check at  $\nu = 0.45$ . The pressure unknown is the mean stress  $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$ , tensile positive, and the  $(2, 2)$  block  $-\mathbf{M}_p/\kappa$  of the Newton system keeps it uniquely determined even when the displacement is prescribed on the whole boundary. The bulk modulus is resolved from  $\nu$  as  $\kappa = 2\mu(1 + \nu)/(3(1 - 2\nu))$  (Section 4.6); at  $\nu = 0.4999$  this gives  $\kappa/\mu \approx 5000$ , the value of the nearly incompressible Cook input. The saddle-point solver of Section 12.4 is built for this regime: its Schur complement approximation  $\mathbf{M}_p(1/\mu + 1/\kappa)$  reduces to the incompressible one as  $\kappa \rightarrow \infty$ .

#### 4.5 Category 4: incompressible

In the limit  $\kappa \rightarrow \infty$  the volumetric energy becomes the internal constraint  $J = 1$  and the pressure a Lagrange multiplier that does no work:

$$\mathbf{P} = \mathbf{P}_{\text{iso}}(\mathbf{F}) + p J \mathbf{F}^{-T}, \quad \int_{\Omega_R} q (J - 1) dV = 0 \quad \forall q. \quad (29)$$

The input selects it with `incompressible: true` or `nu: 0.5`; the materials report  $\kappa = \infty$ , the  $(2, 2)$  block of the Newton system vanishes, and the mixed formulation is required (the displacement formulation refuses the input). With the displacement prescribed on the whole boundary the multiplier is determined only up to a constant (Section 6), so incompressible problems need a traction boundary somewhere. The one exception to the mixed requirement is plane stress (Section 5): for a thin sheet the constraint fixes the thickness stretch  $\lambda_3 = 1/\det \mathbf{F}_{2D}$  and the traction-free face fixes  $p = -\sigma_{\text{iso},33}$  pointwise, so the multiplier is eliminated at the material point and the sheet is a pure displacement problem.

On an incompressible solution the barred and unbarred invariants coincide and the six isochoric energies reduce to the classical incompressible models  $\Psi(I_1, I_2)$  or  $\Psi(\lambda_1, \lambda_2, \lambda_3)$  with  $\lambda_1 \lambda_2 \lambda_3 = 1$ . With  $\Psi_1 = \partial \Psi / \partial I_1$  and  $\Psi_2 = \partial \Psi / \partial I_2$  the Cauchy stress is

$$\boldsymbol{\sigma} = 2\Psi_1 \text{dev } \mathbf{B} - 2\Psi_2 \text{dev } \mathbf{B}^{-1} + p \mathbf{I}, \quad \mu = 2(\Psi_1 + \Psi_2)|_{I_1=I_2=3}, \quad (30)$$

and the principal stress differences  $\sigma_a - \sigma_b = \lambda_a \partial \Psi / \partial \lambda_a - \lambda_b \partial \Psi / \partial \lambda_b$  are constitutive while the mean stress is fixed by equilibrium and the boundary conditions. Much of the literature writes  $\boldsymbol{\sigma} = -\tilde{p} \mathbf{I} + 2\Psi_1 \mathbf{B} - 2\Psi_2 \mathbf{B}^{-1}$  and calls  $\tilde{p}$  the hydrostatic pressure; its value depends on which isotropic terms an author absorbs into it, whereas  $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$  does not, and that is the unknown used here. The closed-form uniaxial, equibiaxial, pure-shear and simple-shear states of these models, for a general  $\Psi(I_1, I_2)$  and specialised to the six energies, serve as reference solutions and are collected in the appendix of the verification manual, which also lists the inputs that reproduce them.

## 4.6 Resolution of the moduli and material regions

The input names a model and its parameters; the factory (`materials.cpp`) resolves the small-strain shear and bulk moduli  $(\mu, \kappa)$  as follows. The coupled models take  $(E, \nu)$  with  $\nu < \frac{1}{2}$ . For a decoupled model  $\mu$  is the model's own small-strain value from the tables above (or  $E/(2(1+\nu))$ ) for the isochoric neo-Hookean model given  $E, \nu$ , and the bulk modulus comes from exactly one of

$$\kappa \text{ given,} \quad \kappa = \frac{2\mu(1+\nu)}{3(1-2\nu)} \text{ from } \nu < \frac{1}{2}, \quad \kappa = \infty \text{ from } \nu = \frac{1}{2} \text{ or } \text{incompressible: true}, \quad (31)$$

which places the material in category 2, 3 or 4. The derived Lamé constant  $\lambda = \kappa - \frac{2}{3}\mu$  is used only in the small-strain checks of the tests.

Materials may vary by element attribute (`material.regions`): every region uses the base model with other parameter values, inheriting the keys it does not give (a region that gives any bulk key replaces the base's bulk specification). The integrators hold a table indexed by attribute and pick the entry per element; all regions must be incompressible or none, and no attribute may be covered twice.

## 5 Two-dimensional problems

### 5.1 Plane strain

By default a two-dimensional mesh is treated as plane strain: the displacement has two components,  $F_{33} = 1$  in (7), and the three-dimensional material returns the full  $\mathbf{P}$ , of which the kernels contract only the in-plane block with  $\text{Grad } \mathbf{w}$ . The out-of-plane stress  $\sigma_{33}$  is a reaction; it is generally non-zero and is included in the pressure, the von Mises stress and the stress outputs. An incompressible plane-strain state has in-plane stretches  $(\lambda, \lambda^{-1})$ .

### 5.2 Plane stress

For a thin sheet loaded in its plane the appropriate reduction is  $\sigma_{i3} = 0$ . The thickness stretch  $\lambda_3 = F_{33}$  is then not prescribed but determined by the material. `plane: stress` (two-dimensional, displacement formulation) wraps the material in the adapter of `plane_stress.hpp`: the kernels still pad  $\mathbf{F}$  with  $F_{33} = 1$ , the adapter replaces  $F_{33}$  by  $\lambda_3$  (and drops the out-of-plane shears) such that  $P_{33} = 0$ , and returns  $\mathbf{P}$  at the completed  $\mathbf{F}_c$ ; its (3,3) entry is zero and the in-plane block is the plane-stress response. Two branches:

**Incompressible base** ( $\kappa = \infty$ ). The constraint fixes the thickness pointwise and  $\sigma_{33} = 0$  fixes the mean stress:

$$\lambda_3 = \frac{1}{\det \mathbf{F}_{2D}}, \quad p = -\sigma_{\text{iso},33} = -\lambda_3 P_{\text{iso},33}, \quad \mathbf{P} = \mathbf{P}_{\text{iso}}(\mathbf{F}_c) + p \mathbf{F}_c^{-T} \quad (J = 1). \quad (32)$$

The Lagrange multiplier is eliminated at the material point: the sheet problem is a pure displacement problem in the two in-plane components with no pressure unknown, no inf-sup condition and no locking. This is the one way to run an incompressible model in the displacement formulation.

**Compressible base.**  $\lambda_3$  solves the scalar equation  $r(\lambda_3) = P_{33}(\mathbf{F}_{2D}, \lambda_3) = 0$  by Newton's method from  $\lambda_3 = 1$ , with  $dr/d\lambda_3$  from a dual evaluation of  $P_{33}$ , steps clipped to  $[-\lambda_3/2, \lambda_3]$  to keep the stretch positive, a relative tolerance of  $10^{-14}$  and at most 100 iterations; a non-finite state (an inverted in-plane block during a line search) returns NaN, which the outer Newton rejects. When the adapter is evaluated with dual numbers the converged root is refined by *one Newton step in dual arithmetic*,

$$\lambda_3^{\text{dual}} = \lambda_3 - \frac{r(\lambda_3)^{\text{dual}} - r(\lambda_3)}{dr/d\lambda_3}, \quad (33)$$

which leaves the value unchanged and gives the derivative part  $-(\partial P_{33}/\partial \mathbf{F} : d\mathbf{F})/(\partial P_{33}/\partial F_{33})$ , the implicit-function derivative  $d\lambda_3/d\mathbf{F}$ . The in-plane tangent obtained by dual seeding is therefore the consistent plane-stress tangent. The energy is the base energy at the completed  $\mathbf{F}_c$ ; since  $P_{33} = 0$  there, its in-plane derivative is the in-plane  $\mathbf{P}$  (envelope theorem). In the small-strain limit the adapter reproduces the plane-stress moduli  $E/(1 - \nu^2)$ ,  $E\nu/(1 - \nu^2)$  and the thickness strain  $-\nu/(1 - \nu) \text{ tr } \boldsymbol{\varepsilon}$ .

Every quadrature quantity uses the completed  $\mathbf{F}_c$  (so `jacobian` is the true  $J$ , one for an incompressible sheet), and `thickness_stretch`  $= \lambda_3$  becomes available. Two cautions: the idealisation assumes stress uniform through the thickness and no bending; and because each load step starts with the boundary displaced and the interior lagging, shear-dominated sheet problems need increments that shear a boundary layer of elements by no more than a few percent, since the energy grows like  $(\det \mathbf{F}_{2D})^{-2}$ .

## 6 Mixed displacement–pressure formulation

For near- and fully incompressible materials the pressure is introduced as an independent field. With  $U(J) = \kappa u(J)$  the material's volumetric law (Section 4.3), the two-field residuals are, for all  $\mathbf{w} \in \mathcal{V}$  and all  $q$  in the pressure space,

$$\begin{aligned} R_u(\mathbf{u}, p; \mathbf{w}) &= \int_{\Omega_R} [\mathbf{P}_{\text{iso}}(\mathbf{F}) + p J \mathbf{F}^{-T}] : \text{Grad } \mathbf{w} \, dV \\ &\quad - \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV - \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA + R_{\text{fol}}(\mathbf{u}; \mathbf{w}) = 0, \\ R_p(\mathbf{u}, p; q) &= \int_{\Omega_R} q \left( u'(J) - \frac{p}{\kappa} \right) dV = 0. \end{aligned} \quad (34)$$

For the default quadratic law  $u'(J) = J - 1$  and the second equation is the familiar  $\int q (J - 1 - p/\kappa) dV = 0$ ; for the quadratic law alone the pair is the stationarity of the perturbed Lagrangian

$$\Pi(\mathbf{u}, p) = \int_{\Omega_R} \left[ \Psi_{\text{iso}}(\mathbf{F}) + p (J - 1) - \frac{p^2}{2\kappa} \right] dV - \Pi_{\text{ext}}(\mathbf{u}). \quad (35)$$

For finite  $\kappa$  the second equation is the weak form of the volumetric law  $p = U'(J)$ , and the mixed and displacement formulations solve the same continuous problem; for  $\kappa = \infty$  it reads  $\int q u'(J) dV = 0$ , the incompressibility constraint  $J = 1$  in weak form for every law, and  $p$  is a Lagrange multiplier. With  $\mathbf{P}_{\text{iso}} : \mathbf{F} = 0$  the Cauchy stress is  $\boldsymbol{\sigma} = J^{-1} \mathbf{P}_{\text{iso}} \mathbf{F}^T + p \mathbf{I}$ , so the unknown  $p$  is the mean stress  $\frac{1}{3} \text{tr } \boldsymbol{\sigma}$ , tensile positive, and the constitutive models need only provide the split of Section 4.3.

**Spaces.** Displacement and pressure are discretised on a Taylor–Hood pair,  $[H^1]^d$  of order  $k$  for  $\mathbf{u}$  and  $H^1$  of order  $k - 1$  for  $p$ , so the input needs  $k \geq 2$ . The affine displacement and the constant pressure lie in these spaces, which is why homogeneous states are reproduced to solver tolerance. Dirichlet conditions act on the displacement only; the pressure carries no essential conditions. With the displacement prescribed on the whole boundary an incompressible pressure is determined only up to a constant, because  $c \int J \mathbf{F}^{-T} : \text{Grad } \mathbf{w} \, dV = c \int_{\partial\Omega_R} \mathbf{w} \cdot J \mathbf{F}^{-T} \mathbf{N} \, dA = 0$  for every admissible  $\mathbf{w}$  (Piola identity); a traction boundary somewhere fixes it.

**Tangent.** The Newton system is the block system

$$\begin{bmatrix} \mathbf{K} & \mathbf{B} \\ \mathbf{B}_{pu} & -\mathbf{M}_p/\kappa \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta p \end{bmatrix} = - \begin{bmatrix} R_u \\ R_p \end{bmatrix}, \quad (36)$$

with, at each quadrature point,

$$\mathbf{K} : \text{Grad } \mathbf{w} : \mathbb{A}(\mathbf{F}, p) : \text{Grad } \Delta \mathbf{u}, \quad \mathbb{A} = \frac{\partial}{\partial \mathbf{F}} [\mathbf{P}_{\text{iso}} + p J \mathbf{F}^{-T}]_{p \text{ fixed}}, \quad (37)$$

$$\mathbf{B} : q(J \mathbf{F}^{-T}) : \text{Grad } \mathbf{w}, \quad \mathbf{B}_{pu} : q u''(J)(J \mathbf{F}^{-T}) : \text{Grad } \Delta \mathbf{u}, \quad \mathbf{M}_p : q \Delta p. \quad (38)$$

The  $(2, 2)$  block vanishes when  $\kappa = \infty$ .  $\mathbb{A}$  at fixed  $p$  is again produced by dual seeding of the combined stress;  $\mathbf{B}$  and  $\mathbf{B}_{pu}$  use  $\partial J / \partial \mathbf{F} = J \mathbf{F}^{-T}$ . For the quadratic law  $u'' = 1$  and  $\mathbf{B}_{pu} = \mathbf{B}^T$ , so the system is symmetric; for the other laws it is not, and the linear solver of Section 12.4 is written for the general case.

## 7 Small-strain linear elasticity

### 7.1 The same weak form

Linearising the kinematics about  $\mathbf{F} = \mathbf{I}$  replaces every strain measure by the infinitesimal strain and removes the distinction between the reference and the current configuration:  $\Omega_R$  is the domain of integration, and  $\mathbf{P}$ ,  $\mathbf{S}$  and  $\boldsymbol{\sigma}$  coincide. With  $\mathbf{H} = \text{Grad } \mathbf{u} = \mathbf{F} - \mathbf{I}$ ,

$$\boldsymbol{\varepsilon} = \text{sym } \mathbf{H}, \quad \boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon} = 2\mu \, \text{dev } \boldsymbol{\varepsilon} + \kappa (\text{tr } \boldsymbol{\varepsilon}) \mathbf{I} = \lambda (\text{tr } \boldsymbol{\varepsilon}) \mathbf{I} + 2\mu \boldsymbol{\varepsilon}, \quad \lambda = \kappa - \frac{2}{3}\mu, \quad (39)$$

with the energy  $\Psi = \mu \, \text{dev } \boldsymbol{\varepsilon} : \text{dev } \boldsymbol{\varepsilon} + \frac{1}{2} \kappa (\text{tr } \boldsymbol{\varepsilon})^2$ . Because  $\boldsymbol{\sigma}$  is symmetric,  $\boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\mathbf{w}) = \boldsymbol{\sigma} : \text{Grad } \mathbf{w}$ , and the weak form of linear elasticity,

$$\int_{\Omega_R} \boldsymbol{\sigma}(\boldsymbol{\varepsilon}(\mathbf{u})) : \boldsymbol{\varepsilon}(\mathbf{w}) \, dV = \int_{\Omega_R} \boldsymbol{\sigma} : \text{Grad } \mathbf{w} \, dV = \int_{\Omega_R} \rho_R \mathbf{b} \cdot \mathbf{w} \, dV + \int_{\Gamma_R^N} \bar{\mathbf{T}} \cdot \mathbf{w} \, dA, \quad (40)$$

is the total Lagrangian form (11) with the flux  $\mathbf{P}(\mathbf{F}) := \boldsymbol{\sigma}(\text{sym}(\mathbf{F} - \mathbf{I}))$ . It is therefore not a new weak form and not a new integrator: `linear_elastic` is a material whose `PK1(F)` returns (39). The element residual is  $\int \sigma_{ij} N_{a,j}$ , the tangent  $\partial \mathbf{P} / \partial \mathbf{F} = \mathbb{C}$ ,  $C_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$ , is constant and has the minor symmetries, and the element matrix  $\int N_{a,j} C_{ijkl} N_{b,l}$  is the classical  $\mathbf{B}^T \mathbb{C} \mathbf{B}$  stiffness (it equals MFEM's `ElasticityIntegrator` to round-off; the tangents of the compressible hyperelastic models at  $\mathbf{u} = \mathbf{0}$  are this matrix). The model is not objective under finite rotations, by construction; it is invariant under infinitesimal ones,  $\mathbf{P}(\mathbf{I} + \mathbf{W}) = \mathbf{0}$  for skew  $\mathbf{W}$ . The input keys are those of `iso_neo_hookean`: `mu`, or `E` and `nu`, and the bulk modulus from exactly one of `kappa`, `nu`, `incompressible`; there is no volumetric law to choose.

## 7.2 What depends on the kinematics

The flux contract of the kernels is the same for both kinematics. What differs lies outside the flux, and is switched by a compile-time trait of the material (`materials/kinematics.hpp`; the plane-stress adapter inherits the trait of its base):

|                                                   | Finite strain                                                                                         | Small strain                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Cauchy stress (outputs, von Mises)                | $\boldsymbol{\sigma} = J^{-1} \mathbf{P} \mathbf{F}^T$                                                | $\boldsymbol{\sigma} = \mathbf{P}$                                                                               |
| Volume ratio (jacobian)                           | $J = \det \mathbf{F}$                                                                                 | $1 + \text{tr } \boldsymbol{\varepsilon}$                                                                        |
| strain                                            | Green–Lagrange<br>$\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$                    | $\boldsymbol{\varepsilon} = \text{sym}(\mathbf{F} - \mathbf{I})$                                                 |
| Volume measure of the mixed form and its gradient | $J, \partial J / \partial \mathbf{F} = J \mathbf{F}^{-T}$                                             | $\theta = 1 + \text{tr } \boldsymbol{\varepsilon}, \partial \theta / \partial \mathbf{F} = \mathbf{I}$           |
| Incompressible plane stress                       | $\lambda_3 = 1 / \det \mathbf{F}_{2D},$<br>$\mathbf{P} = \mathbf{P}_{\text{iso}} + p \mathbf{F}^{-T}$ | $\varepsilon_{33} = -(\varepsilon_{11} + \varepsilon_{22}), \mathbf{P} = \mathbf{P}_{\text{iso}} + p \mathbf{I}$ |
| Follower pressure                                 | a load of its own (Section 9.4)                                                                       | identical to the dead pressure;<br>rejected as an input error                                                    |
| Moment arms of the reactions                      | current positions $\mathbf{X} + \mathbf{u}$                                                           | reference positions $\mathbf{X}$                                                                                 |

A stress formed with the finite-strain push-forward would be wrong by  $O(|\text{Grad } \mathbf{u}|)$  and not symmetric; with reference moment arms the moment balance of the reactions closes exactly, as the equilibrium equations of the linear theory are written on the reference configuration. The outputs `pk1_stress` and `cauchy_stress` coincide, the `deformation_gradient` is  $\mathbf{I} + \text{Grad } \mathbf{u}$ , and the `thickness_stretch` is  $1 + \varepsilon_{33}$ .

## 7.3 Two-dimensional problems

Plane strain is the padded  $F_{33} = 1$ , i.e.  $\varepsilon_{33} = 0$ , with  $\sigma_{33} = \lambda(\varepsilon_{11} + \varepsilon_{22})$  carried in the outputs. Under plane stress the compressible branch of the adapter of Section 5 applies unchanged: its scalar Newton iteration on  $P_{33} = 0$  is exact after one step and returns  $\varepsilon_{33} = -\frac{\nu}{1-\nu}(\varepsilon_{11} + \varepsilon_{22})$ , and the dual refinement returns the plane-stress moduli,  $\sigma_{\alpha\beta} = \lambda^* \varepsilon_{\gamma\gamma} \delta_{\alpha\beta} + 2\mu \varepsilon_{\alpha\beta}$  with  $\lambda^* = 2\lambda\mu/(\lambda + 2\mu)$ , that is  $E/(1 - \nu^2)$  and  $E\nu/(1 - \nu^2)$ . For  $\nu = \frac{1}{2}$  the thickness strain and the mean stress are eliminated in closed form,  $\lambda^* = 2\mu$ , and no pressure unknown is needed.

## 7.4 Mixed formulation (Herrmann)

With the volume measure  $\theta = 1 + \text{tr } \boldsymbol{\varepsilon}$  and  $\partial \theta / \partial \mathbf{F} = \mathbf{I}$  in place of  $J$  and  $J \mathbf{F}^{-T}$ , the residuals (34) become

$$R_u = \int_{\Omega_R} [2\mu \text{dev } \boldsymbol{\varepsilon} + p \mathbf{I}] : \text{Grad } \mathbf{w} \, dV - \text{loads}, \quad R_p = \int_{\Omega_R} q \left( \text{tr } \boldsymbol{\varepsilon} - \frac{p}{\kappa} \right) dV, \quad (41)$$

Herrmann’s formulation of linear elasticity [10]: the material provides  $\mathbf{P}_{\text{iso}} = 2\mu \text{dev } \boldsymbol{\varepsilon}$  and the volumetric law  $u(\theta) = \frac{1}{2}(\theta - 1)^2$ , and  $\kappa = \infty$  gives the incompressible problem  $\text{tr } \boldsymbol{\varepsilon} = 0$  in weak form. The blocks of (36) are  $\mathbf{B} : q \text{div } \mathbf{w}$ ,  $\mathbf{B}_{pu} = \mathbf{B}^T$ , and  $\mathbf{K}$  carries no pressure term, so the system is symmetric, independent of the state, and  $\mathbf{K}$  is positive semi-definite; the saddle-point solver of Section 12.4 applies unchanged. As for the finite-strain materials the constraint holds weakly: the pointwise  $\text{tr } \boldsymbol{\varepsilon}$  of an incompressible solution is  $O(h^2)$ , not zero.

## 7.5 Solution of a linear problem

The residual is affine in the unknown, so the first Newton step is the exact solve: one assembly, one preconditioner setup, one Krylov solve. Four points follow.

- *The residual has a round-off floor that can lie above the Newton tolerance.* The residual  $\mathbf{K}\mathbf{u} - \mathbf{f}$  is evaluated with an error of the order of  $\epsilon_{\text{mach}}\|\mathbf{K}\|\|\mathbf{u}\|$ , that is  $\epsilon_{\text{mach}}\|\mathbf{K}\|\|\mathbf{u}\|/\|\mathbf{f}\|$  relative to  $\|\mathbf{R}_0\|$ . Bending makes this ratio large: measured values are  $4 \times 10^{-11}$  for Cook’s membrane,  $8 \times 10^{-11}$  for a 10:1 cantilever and  $1.6 \times 10^{-9}$  for a 20:1 beam, and tightening the Krylov tolerance does not move them. A nonlinear problem passes such a tolerance by quadratic convergence; a linear one lands on the floor and stays there, and a second step then fails its line search. A problem therefore declares itself linear (`QuasiStaticProblem::IsLinear`: a small-strain material with dead loads), and for such a problem a second or later Newton step that fails its line search, or reduces the residual by less than a factor of ten, is taken as having met the floor: the state is accepted (`NewtonReport::at_floor`), provided every linear solve converged. Steps that still reduce the residual tenfold (an inexact linear solver, i.e. iterative refinement) continue as usual, and nothing changes for a nonlinear problem.
- *Tolerances.* `newton.rtol` only accepts the linear solve; the accuracy of the solution is set by `linear.rtol`. The inputs use  $10^{-8}$  and  $10^{-13}$ . The mixed formulation typically takes a second step, which refines the first.
- *The tangent predictor is excluded.* After an exact predictor Newton would start at the floor, where `rtol`  $\|\mathbf{R}_0\|$  is meaningless; `solver.predictor: tangent` is an input error for a small-strain material.
- *The Jacobian is assembled once for a load path.* It depends neither on the state nor on the pseudo-time (the essential degrees of freedom are fixed when the loads are finalised), so a problem that declares itself linear returns the same assembled matrix from every Jacobian request until its boundary conditions change. The physics shares an *operator stamp* with the linear solver it creates, a counter incremented at every assembly; a solver that is handed the same operator with an unchanged stamp keeps its setup, the AMG hierarchy and, in the mixed formulation, the augmented blocks of Section 12.4. The address of the matrix alone would not identify it, since a reassembled matrix can occupy the address of the old one. A nonlinear problem stamps at every request, so nothing changes for it. The results are identical bit for bit to reassembling; over ten steps the time spent in assembly and setup drops by a factor of seven to nine. The Krylov solves remain and dominate these runs, so the overall gain is about 1.5 for the displacement formulation and negligible for the mixed one.

Forming  $\mathbf{F} = \mathbf{I} + \mathbf{H}$  and subtracting  $\mathbf{I}$  again leaves an absolute error of  $\epsilon_{\text{mach}}$  in  $\boldsymbol{\varepsilon}$ , a relative floor of  $\epsilon_{\text{mach}}/|\text{Grad } \mathbf{u}|$  in the stress. Since the solution scales with the load, inputs choose amplitudes with  $|\text{Grad } \mathbf{u}| \gtrsim 10^{-4}$  and scale the results, never the reverse; the “linear limit” runs of the hyperelastic models, which must keep  $|\text{Grad } \mathbf{u}|$  small, do not have that freedom.

## 8 Elastodynamics

A `dynamics` block in the input adds the inertial term to either weak form. Without it nothing of this section exists: the quasi-static analysis is unchanged to the last digit.

## 8.1 Strong and weak form; the constant mass matrix

With  $t$  now the physical time, balance of linear momentum on the reference configuration and its weak form are

$$\rho_R \ddot{\mathbf{u}} = \text{Div } \mathbf{P} + \rho_R \mathbf{b} \text{ in } \Omega_R, \quad \int_{\Omega_R} \rho_R \ddot{\mathbf{u}} \cdot \mathbf{w} \, dV + R(\mathbf{u}; \mathbf{w}) = 0 \quad \forall \mathbf{w} \in \mathcal{V}, \quad (42)$$

with  $R$  the quasi-static residual (11) (internal minus external) and initial data  $\mathbf{u}(\cdot, 0) = \mathbf{u}_0$ ,  $\dot{\mathbf{u}}(\cdot, 0) = \mathbf{v}_0$ . In total Lagrangian form the inertial term is integrated over the *fixed* mesh with the *reference* density: mass conservation,  $\rho_R = J\rho$ , is built in, and the mass matrix

$$M_{ai,bj} = \delta_{ij} \int_{\Omega_R} \rho_R N_a N_b \, dV \quad (43)$$

is constant. It is assembled once, the same for finite and small strain and for every material; it is the consistent mass, with  $\rho_R$  piecewise constant by element attribute (`material.rho0` and the regions'), the same coefficient that multiplies the body force. In two dimensions it is the mass per unit reference thickness, also under plane stress, where the current thickness changes and the mass does not. The semi-discrete system is

$$\mathbf{M} \ddot{\mathbf{u}} + \mathbf{S}(\mathbf{u}, t) = \mathbf{0}, \quad (44)$$

where  $\mathbf{S}$  is the static residual vector of the wrapped problem (its `Mult`) and  $\mathbf{K} = \partial \mathbf{S} / \partial \mathbf{u}$  its Jacobian.

## 8.2 Time integration in displacement form

One family of implicit one-step schemes is implemented (`solvers/time_integration.hpp`), with the displacement of the new time level as the unknown, so that prescribed displacements, Newton, the line search and the linear solvers are those of the quasi-static analysis. With  $(\mathbf{u}_n, \mathbf{v}_n, \mathbf{a}_n)$  known and  $\Delta t = t_{n+1} - t_n$ ,

$$\begin{aligned} \mathbf{u}^* &= \mathbf{u}_n + \Delta t \mathbf{v}_n + \Delta t^2 \left(\frac{1}{2} - \beta\right) \mathbf{a}_n, & \mathbf{a}(\mathbf{u}) &= \frac{\mathbf{u} - \mathbf{u}^*}{\beta \Delta t^2}, \\ \mathbf{v}^* &= \mathbf{v}_n + \Delta t(1 - \gamma) \mathbf{a}_n, & \mathbf{v}(\mathbf{u}) &= \mathbf{v}^* + \gamma \Delta t \mathbf{a}(\mathbf{u}), \end{aligned} \quad (45)$$

the balance is imposed between the time levels with *interpolated forces* (the form of the HHT operator of general-purpose codes),

$$\mathbf{M}[(1 - \alpha_m) \mathbf{a}(\mathbf{u}) + \alpha_m \mathbf{a}_n] + (1 - \alpha_f) \mathbf{S}(\mathbf{u}, t_{n+1}) + \alpha_f \mathbf{S}_n = \mathbf{0}, \quad \mathbf{S}_n = \mathbf{S}(\mathbf{u}_n, t_n). \quad (46)$$

Interpolating the forces rather than the state means  $\mathbf{S}$  is only ever evaluated at  $(\mathbf{u}_{n+1}, t_{n+1})$ : the load set, the Dirichlet data and the follower loads keep one time. The parameter sets are

| scheme            | $\alpha_m$                                 | $\alpha_f$                            | $\beta$                                  | $\gamma$                            |
|-------------------|--------------------------------------------|---------------------------------------|------------------------------------------|-------------------------------------|
| newmark           | 0                                          | 0                                     | given ( $\frac{1}{4}$ )                  | given ( $\frac{1}{2}$ )             |
| hht               | 0                                          | $\alpha \in [0, \frac{1}{3}]$         | $\frac{1}{4}(1 + \alpha)^2$              | $\frac{1}{2} + \alpha$              |
| generalized_alpha | $\frac{2\rho_\infty - 1}{\rho_\infty + 1}$ | $\frac{\rho_\infty}{\rho_\infty + 1}$ | $\frac{1}{4}(1 - \alpha_m + \alpha_f)^2$ | $\frac{1}{2} - \alpha_m + \alpha_f$ |

(Newmark [12]; Hilber, Hughes and Taylor [13]; Chung and Hulbert [14]). All are second-order accurate with  $\gamma = \frac{1}{2} - \alpha_m + \alpha_f$  and, for a linear problem, unconditionally stable; **newmark** with  $\gamma > \frac{1}{2}$  is first order, and with  $2\beta < \gamma$  only conditionally stable (the run header says so).  $\rho_\infty \in [0, 1]$  is the spectral radius of the amplification matrix at infinite frequency: 1 means no numerical dissipation, 0 annihilates in a few steps whatever the step does not resolve. The default **newmark** pair  $(\frac{1}{4}, \frac{1}{2})$  is the trapezoidal rule, which for a linear problem conserves the energy exactly and lengthens a period by the factor  $(\omega\Delta t/2)/\arctan(\omega\Delta t/2) \approx 1 + (\omega\Delta t)^2/12$ . For a nonlinear problem it is not unconditionally stable, and what a mesh cannot resolve behind a wave front rings for ever; **generalized\_alpha** with  $\rho_\infty$  between 0.8 and 0.9 removes the unresolved part at little cost to the resolved one. It dissipates in a norm of its own, not step by step in the physical energy, which may rise slightly between two steps but does not exceed its initial value.

**The step equation.** Divided by  $1 - \alpha_f$ , (46) reads

$$\boxed{\mathbf{G}(\mathbf{u}) = \mathbf{S}(\mathbf{u}, t_{n+1}) + c_M \mathbf{M}(\mathbf{u} - \mathbf{u}^*) + \mathbf{h}_n = \mathbf{0}, \quad \frac{\partial \mathbf{G}}{\partial \mathbf{u}} = \mathbf{K}(\mathbf{u}) + c_M \mathbf{M},} \quad (47)$$

$$c_M = \frac{1 - \alpha_m}{(1 - \alpha_f) \beta \Delta t^2}, \quad \mathbf{h}_n = \frac{\alpha_m \mathbf{M} \mathbf{a}_n + \alpha_f \mathbf{S}_n}{1 - \alpha_f}. \quad (48)$$

The static operator enters with unit weight: inertia is a linear spring  $c_M \mathbf{M}$  plus a known history load. This is why it needs no kernel of its own. `DynamicSolidProblem` (`physics/dynamic_solid_problem.hpp`) wraps any solid problem and *is* equation (47) behind the interface the stepper already drives; a linear problem stays linear (the floor logic of Section 7.5 applies unchanged), and the consistent Jacobian  $\mathbf{K} + c_M \mathbf{M}$  keeps Newton quadratic (measured order 1.9 at displacements of 0.75 of the body's size). After convergence,  $\mathbf{a}_{n+1} = \mathbf{a}(\mathbf{u}_{n+1})$  and  $\mathbf{v}_{n+1} = \mathbf{v}(\mathbf{u}_{n+1})$ .  $\mathbf{M}$  is kept twice: whole, for the residual, the energies and the reactions (it couples the acceleration of a prescribed boundary into the free rows), and with the essential rows and columns zeroed for the Jacobian.

**Initial state.**  $\mathbf{u}_0$  and  $\mathbf{v}_0$  are given as expressions of the reference coordinates (zero by default); the Dirichlet data of  $t = 0$  overwrite  $\mathbf{u}_0$  on the prescribed degrees of freedom, with a warning if that changes it. The initial acceleration is the consistent one,  $\mathbf{M} \mathbf{a}_0 = -\mathbf{S}(\mathbf{u}_0, 0)$  on the free degrees of freedom and zero on the prescribed ones, with the loads at  $t = 0$  taken as their *right limit*: a load that is switched on at  $t = 0$  and held is on from the first instant and enters  $\mathbf{a}_0$ .

**Prescribed motion and reactions.** Dirichlet data are imposed on  $\mathbf{u}_{n+1}$ ; velocity and acceleration of those degrees of freedom follow from the update formulas like everywhere else. The reaction of a Dirichlet entry is the full balance  $\mathbf{S}_{\text{full}}(\mathbf{u}_{n+1}, t_{n+1}) + \mathbf{M} \mathbf{a}_{n+1}$  on its rows: a support force includes the inertia it carries.

**Energy balance.** The kinetic energy is  $\frac{1}{2} \mathbf{v} \cdot \mathbf{M} \mathbf{v}$ , the internal energy that of Section 13, and the external work is accumulated by the trapezoidal rule over the dead loads and over the supports,  $\frac{1}{2}(\mathbf{f}_n + \mathbf{f}_{n+1} + \mathbf{r}_n + \mathbf{r}_{n+1}) \cdot (\mathbf{u}_{n+1} - \mathbf{u}_n)$  with  $\mathbf{r}$  the reactions on the prescribed rows. For a linear problem and the trapezoidal rule, kinetic plus internal energy minus external work is constant *exactly* (it is an algebraic identity of the scheme), which makes the balance a test of the implementation and not only an output; for a nonlinear problem its error is  $O(\Delta t^2)$ . Follower pressures are not dead loads and are not counted.

**A linear problem.**  $\mathbf{K} + c_M \mathbf{M}$  is constant while  $\Delta t$  is, so it is formed, and the AMG hierarchy built, once per run (the operator stamp of Section 7.5, now of the sum). Equal steps  $t_{\text{final}}k/n$  differ in the last digits of  $t_{k+1} - t_k$ ; the increment of the previous step is kept when the new one agrees with it to  $10^{-12}$ , which also makes a run independent of how its time grid was formed. Such a run is bound by its residual evaluations (three per step against one Krylov solve); the full static residual that reactions and the external work need is evaluated only when one of them is asked for.

### 8.3 Mixed formulation: a differential-algebraic system

In the  $u$ - $p$  formulation the pressure carries no inertia:  $\mathbf{M}$  acts on the displacement block, the constraint row is that of the static problem at  $t_{n+1}$ , unweighted, and  $\mathbf{h}_n$  has no pressure block. The Jacobian is that of Section 6 with  $\mathbf{K} + c_M \mathbf{M}$  in place of  $\mathbf{K}$ . Two properties follow.

- *The pressure mode.* The pressure force is part of  $\mathbf{S}_u$  and is interpolated with it, so an error  $e_n$  in  $p_n$  returns as  $e_{n+1} = -\alpha_f/(1 - \alpha_f) e_n = -\rho_\infty e_n$ : a sign-alternating mode that decays only for  $\rho_\infty < 1$  (measured: two runs that differ in  $p_0$  alone differ at every later step by 0.6000 times the step before, for  $\rho_\infty = 0.6$ ). The trapezoidal rule has the same factor 1 through the acceleration. A mixed dynamic analysis therefore uses a dissipative scheme, and the run header warns otherwise. The initial pressure is that of  $\mathbf{u}_0$  at finite  $\kappa$  (one solve with the constraint row) and zero for an incompressible material, whose true initial pressure would follow from a constraint on the acceleration that is not enforced; the start-up error decays with the same factor.
- *The pressure carries the tolerance of the solves multiplied by  $c_M$ .* It balances  $\mathbf{M}\mathbf{a}$ , and  $\mathbf{a} = (\mathbf{u} - \mathbf{u}^*)/(\beta \Delta t^2)$  amplifies the error of a solve by  $O(\Delta t^{-2})$ . With `newton.rtol`  $10^{-10}$  this noise, not the time integration, limits the pressure at small steps; tighten the tolerances when the pressure matters.

With  $u$  and  $p$  held exactly by the spaces, both are second order in  $\Delta t$ , for  $\nu = 0.4999$  and for  $\kappa = \infty$ . The linear solver is extended for the mass-dominated displacement block (Section 12.4).

## 9 Boundary conditions and loading

All boundary conditions and the body force are entries of a *load set* (`physics/loads.hpp`) attached to the displacement space and shared by both formulations. Each entry  $i$  has data (an MFEM coefficient built from the input expressions) and a scalar schedule  $s_i(t)$  of the pseudo-time.

### 9.1 Pseudo-time and schedules

The load path is parametrised by  $t \in [0, 1]$ . The data of entry  $i$  enters as  $s_i(t) f_i(\mathbf{X}, t)$ , where  $f_i$  may itself depend on  $t$  through its expression. Schedules are piecewise linear:

$$s(t) = \begin{cases} \text{ramp}(t_0, t_1) : & 0 \ (t \leq t_0), \ \frac{t - t_0}{t_1 - t_0} \ (t_0 < t < t_1), \ 1 \ (t \geq t_1), \\ \text{constant} : & 1 \ (t > 0), \\ \text{table}(t_k, s_k) : & \text{linear interpolation of } (t_k, s_k), \text{ clamped outside.} \end{cases} \quad (49)$$

The default is the ramp over  $[0, 1]$ , unless the expression mentions  $t$ , in which case it is the constant schedule and  $t$  enters through the function only; a schedule given together with such an expression multiplies it. With the ramp on every entry,  $t < 1$  solves a proportionally scaled problem and  $t = 1$  the problem of the weak form. Since hyperelasticity without follower loads is conservative, the

converged state at  $t = 1$  does not depend on the path; staged loading is a test of the implementation, not a different problem. Follower pressures are not conservative and can produce path dependence in the presence of instabilities.

**Physical time.** In a dynamic analysis (Section 8)  $t$  is the physical time in  $[0, t_{\text{final}}]$ . Ramps and tables are given on that interval (a ramp without bounds spans it), the value at  $t = 0$  is the right limit, so the constant schedule is 1 for  $t \geq 0$ , and an entry *without* a schedule is constant: its data is the expression as it stands, a step load when the expression does not mention  $t$ . The quasi-static default ramp over the pseudo-time has no meaning there. Because that default depends on the analysis, the executable prints the resolved time dependence of every entry in the header of a dynamic run.

## 9.2 Prescribed displacements

A Dirichlet entry prescribes, on a set of boundary attributes, either all components or the listed subset (**components**), which is how rollers and symmetry planes are expressed. The essential true degrees of freedom of each entry are collected per component; their union is the essential set of the problem. Before each Newton solve the stepper overwrites the essential entries of the solution vector with  $s_i(t)$  times the boundary projection of  $\bar{\mathbf{u}}_i$  (**ApplyDirichlet**); the residual is zeroed and the Jacobian rows and columns eliminated there. Two entries may overlap on shared edges or corners; the later entry wins and a warning names entries that prescribe the same component on the same attribute.

## 9.3 Dead loads

Nominal tractions  $\bar{\mathbf{T}}$  (per unit reference area), dead normal pressures  $\bar{\mathbf{T}} = -p\mathbf{N}$  on the reference outward normal, and the body force  $\rho_R \mathbf{b}$  are assembled once each into true-dof vectors  $L_i$  with stock MFEM linear-form integrators; entries whose expression depends on  $t$  are reassembled whenever  $t$  changes. The external load at time  $t$  is  $\sum_i s_i(t) L_i$  and is subtracted from the internal force in the residual.

## 9.4 Follower pressure

A pressure  $p$  per unit *current* area on  $\Gamma_R^p$  exerts the spatial traction  $-p\mathbf{n}$ ; by Nanson's formula (6) its nominal counterpart is  $\bar{\mathbf{T}}(\mathbf{u}) = -p J \mathbf{F}^{-T} \mathbf{N} = -p \operatorname{cof} \mathbf{F} \mathbf{N}$ , which depends on the displacement. It is therefore a boundary term of the nonlinear form,

$$R_{\text{fol}}(\mathbf{u}; \mathbf{w}) = \sum_i s_i(t) \int_{\Gamma_R^{p_i}} p_i (\operatorname{cof} \mathbf{F} \mathbf{N}) \cdot \mathbf{w} \, dA, \quad (50)$$

with the load stiffness

$$D_{\Delta \mathbf{u}} \int_{\Gamma_R^p} p (\operatorname{cof} \mathbf{F} \mathbf{N}) \cdot \mathbf{w} \, dA = \int_{\Gamma_R^p} p [(\mathbf{F}^{-T} : \operatorname{Grad} \Delta \mathbf{u}) J \mathbf{F}^{-T} \mathbf{N} - J \mathbf{F}^{-T} (\operatorname{Grad} \Delta \mathbf{u})^T \mathbf{F}^{-T} \mathbf{N}] \cdot \mathbf{w} \, dA, \quad (51)$$

using  $DJ = J \mathbf{F}^{-T} : \operatorname{Grad} \Delta \mathbf{u}$  and  $D\mathbf{F}^{-T} = -\mathbf{F}^{-T} (\operatorname{Grad} \Delta \mathbf{u})^T \mathbf{F}^{-T}$ . This tangent is not symmetric, so the conjugate gradient option of the linear solver is refused for inputs with a follower pressure. The dead pressure and the follower pressure coincide at  $\mathbf{F} = \mathbf{I}$  and differ at  $O(p^2)$ . The implementation (**kernels/follower\_pressure.hpp**) is described in Section 10.4.

## 9.5 Reactions

The reaction of a Dirichlet entry is the resultant of the nodal forces the constraint exerts on the body: the residual (11) evaluated with the essential rows *not* zeroed, restricted to the entry's essential degrees of freedom (internal minus external nodal forces). This is the exact discrete counterpart of the traction integral over the constrained face. The moment is taken about the origin at the current nodal positions  $\mathbf{X} + \mathbf{u}$ . Two entries that share nodes both count the shared forces.

## 9.6 Expressions

Boundary data and the body force are strings  $f(x, y, z, t)$  in the reference coordinates and the pseudo-time ( $z = 0$  in two dimensions), parsed once into a postfix program and evaluated per point without allocation. The grammar has numbers, `x y z t, pi`, the operators `+ - * / ^` (`^` binds tightest, associates to the right, and unary minus binds looser, so `-x^2 = -(x^2)`), parentheses, the functions `sin cos tan exp log sqrt abs pow min max`, and `if(cond, a, b)` with the comparisons `< <= > >= == !=` evaluating to 1 or 0. Expressions cannot refer to the solution; a load that depends on the deformation is a follower load, and only the follower pressure is provided.

# 10 Finite element discretisation

## 10.1 Spaces and quadrature

The displacement lives in the vector  $H^1$  space of polynomial order  $k$  (`mesh.order`) on the reference mesh, ordered by vector dimension (the components of a node are consecutive), so that true-dof vectors can be read node by node; the pressure of the mixed formulation in the scalar  $H^1$  space of order  $k - 1$ . Quadrilaterals, triangles, hexahedra and tetrahedra, including mixed and curved (second-order geometry) meshes, are accepted. Volume and face integrals use the Gauss rule of order  $2k + 3$  on the element geometry; the same rule defines the quadrature space of the post-processed quantities (Section 13).

## 10.2 Element residual and tangent of the displacement formulation

At a quadrature point with weight  $w_q$  (the rule weight times the reference map's Jacobian determinant) the physical shape-function gradients  $D_{aj} = \partial N_a / \partial X_j$  give  $H_{ij} = \sum_a u_{ai} D_{aj}$  and the material returns  $\mathbf{P}$  and  $\mathbb{A}$  at  $\mathbf{F} = \mathbf{I} + \mathbf{H}$ . The element residual and tangent are

$$r_{ai} = \sum_q w_q P_{ij} D_{aj}, \quad K_{(ai),(bk)} = \sum_q w_q D_{aj} A_{ijkl} D_{bl}, \quad (52)$$

summed over the in-plane indices  $j, l < d$  (in two dimensions the out-of-plane components of  $\mathbb{A}$  are never used). The loops are written in `kernels/total_lagrangian.hpp` as plain host code without allocation or virtual calls; the quadrature-point bodies (deformation gradient, stress, tangent, Cauchy stress) are free functions on plain tensors. The integrator is a template over the material type, instantiated once per model through `std::visit` at setup, so no dispatch happens per quadrature point. The energy  $\int \Psi dV$  is provided by the same integrator when the material has an energy, and is used only as a diagnostic.

### 10.3 Element blocks of the mixed formulation

The block integrator (`kernels/mixed_total_lagrangian.hpp`) adds the pressure shape functions  $\mathbf{N}_b$  and forms, per quadrature point,

$$\begin{aligned} r_{ai}^u &= w_q [P_{\text{iso},ij} + p(J\mathbf{F}^{-T})_{ij}] D_{aj}, & r_b^p &= w_q \left( u'(J) - \frac{p}{\kappa} \right) \mathbf{N}_b, \\ K_{(ai),(bk)}^{uu} &= w_q D_{aj} A_{ijkl} D_{bl}, & K_{(ai),b}^{up} &= w_q (J\mathbf{F}^{-T})_{ij} D_{aj} \mathbf{N}_b, \\ K_{b,(ai)}^{pu} &= u''(J) K_{(ai),b}^{up}, & K_{ab}^{pp} &= -\frac{w_q}{\kappa} \mathbf{N}_a \mathbf{N}_b, \end{aligned} \quad (53)$$

with  $p = \sum_b p_b \mathbf{N}_b$ ,  $\mathbb{A}$  the dual tangent of  $\mathbf{P}_{\text{iso}} + pJ\mathbf{F}^{-T}$  at fixed  $p$ , and  $u', u''$  the normalised derivatives of the volumetric law. In two dimensions  $J$  and  $J\mathbf{F}^{-T}$  are those of the  $2 \times 2$  block (plane strain). The energy of this integrator is the mixed functional  $\Psi_{\text{iso}} + p(J - 1) - \kappa u^*(p/\kappa)$ , with  $u^*$  the Legendre transform of the volumetric law (Section 13); the last term is absent when  $\kappa = \infty$ .

### 10.4 Follower-pressure face kernel

On each boundary face of  $\Gamma_R^p$  the integrator takes the adjacent element's displacement degrees of freedom, evaluates  $\mathbf{H}$  and  $\mathbf{F} = \mathbf{I} + \mathbf{H}$  at the face quadrature points, and obtains  $\mathbf{N} dA$  as the area-weighted reference normal of the face map (`CalcOrtho`), oriented outward by a one-time check against the vector from the element centre to the face centre. The face residual is  $r_{ai} = \sum_q s \hat{w}_q p N_a (\text{cof } \mathbf{F} \mathbf{N})_i$  with  $\hat{w}_q$  the face rule weight; the tangent is obtained by seeding  $\mathbf{F}$  with dual numbers in the direction of each element displacement degree of freedom ( $\Delta u_k = N_b \Rightarrow dF_{kj} = D_{bj}$ ) and reading the derivative part of  $\text{cof } \mathbf{F} \mathbf{N}$ , which reproduces (51) without writing it out. The scale  $s$  is the schedule value  $s_i(t)$ , owned by the load set and read through a pointer so that the integrator needs no rebuild when  $t$  changes. In the mixed formulation the same kernel acts on the displacement block only.

### 10.5 The assembled operator

Each physics module is an `mfem::Operator` on true degrees of freedom: `Mult` returns the residual (11) or (34) with the essential rows zeroed; `GetGradient` returns the assembled `HypreParMatrix` (or the  $2 \times 2$  `BlockOperator` of such matrices) with the essential rows and columns eliminated. Full assembly through `ParNonlinearForm` and `ParBlockNonlinearForm` is used; partial assembly is not implemented.

## 11 Automatic differentiation of the tangents

A dual number  $a = (a_v, a_d)$  carries a value and one directional derivative with the arithmetic  $(a + b) = (a_v + b_v, a_d + b_d)$ ,  $(ab) = (a_v b_v, a_d b_v + a_v b_d)$ , and the chain rule for  $/$ ,  $\log$ ,  $\exp$ ,  $\sqrt{\cdot}$ ,  $\text{pow}$ ,  $|\cdot|$ ,  $\sin$ ,  $\cos$  (`base/dual.hpp`). The fixed-size tensors of `base/tensor.hpp` are templated on the scalar type, so any material written with their operations compiles for  $T = \text{dual}$  unchanged.

The material tangent (13) is obtained by seeding: for each in-plane index pair  $(k, l)$  the entry  $F_{kl}$  is given derivative part one,  $\mathbf{P}$  is evaluated once with duals, and its derivative parts fill  $A_{ijkl}$ . This costs  $d^2$  stress evaluations per quadrature point (nine in three dimensions, four in plane strain) and produces the exact tangent of the implemented stress to round-off, with no hand-coded derivatives. The same seeding gives the tangent of  $\mathbf{P}_{\text{iso}} + pJ\mathbf{F}^{-T}$  at fixed  $p$  in the mixed kernel and the follower

load stiffness. Two places need more than plain forward mode: the Ogden model, whose eigenvectors are not differentiable at coincident stretches (Section 4.3.3), and the compressible plane-stress adapter, whose thickness stretch is the root of a scalar equation (Section 5); both supply the derivative part analytically or implicitly inside their dual overloads, so the seeding loop is unchanged. All tangents are checked against central finite differences in the tests.

## 12 Solution algorithms

### 12.1 Quasi-static load stepping

The stepper (`solvers/quasi_static.hpp`) advances  $t$  through a sequence of breakpoints  $0 < t_1 < \dots < t_n = 1$  (equal increments from `load_steps`, or the segments of `steps`). For each increment it sets the pseudo-time of the load set (schedules evaluated, coefficients' time set, time-dependent dead loads reassembled, follower scales updated), applies the Dirichlet data to the current solution vector, and runs Newton warm-started from the last converged state. If Newton fails and `substep.on_failure` is set, the last converged state is restored, the increment is halved and retried, at most `max_bisections` times and never below `min_dt`; after a converged reduced increment the stepper aims at the planned breakpoint again and never grows beyond the planned grid. Each accepted increment triggers the output callback (fields, ParaView cycle, probes, reactions).

**Time stepping.** A dynamic analysis is advanced by the same loop (`SolveDynamic`) over targets  $0 < t_1 < \dots < t_n = t_{\text{final}}$  from `dynamics.dt` or the segments of `dynamics.steps`. The problem is then the step equation (47): setting the time begins a step (increment,  $c_M$ ,  $\mathbf{u}^*$ ,  $\mathbf{h}_n$ , the time of the loads), and an accept hook, called once per accepted increment, advances the history ( $\mathbf{u}_n, \mathbf{v}_n, \mathbf{a}_n, \mathbf{S}_n$ ). A rejected increment leaves the history untouched, so bisection needs nothing else; a run with a halved step equals the run with the two half steps planned. Two details differ from the pseudo-time. A target counts as reached within  $10^{-9}$  of the planned increment, not within an absolute  $10^{-14}$ , which is below the spacing of floating-point numbers once  $t > 100$  and would leave a degenerate last step with  $c_M \sim \Delta t^{-2}$ . And Newton starts from the last converged state with the new Dirichlet data, never from an extrapolation in time: a better start makes  $\|R_0\|$  small, and `rtol`  $\|R_0\|$  then lies below the round-off floor of the residual, the trap that excludes the tangent predictor for linear problems. `substep.min_dt` is a time in a dynamic analysis ( $10^{-3}$  of the smallest planned step by default); `load_steps`, `steps` and `predictor` of the solver section are input errors.

**Initial guess of an increment.** By default the increment starts from the last converged state  $x_n$  with the new Dirichlet values written on the boundary. The interior lags, so the first Newton linearisation is about a state whose boundary layer of elements carries the whole increment; in a sheared block the iterate that follows can have  $J$  between 0.4 and 3 at quadrature points, and the next step can invert elements. With `solver.predictor: tangent` the stepper instead imposes the increment on a linear solve about the converged state. With  $g$  the state  $x_n$  carrying the new boundary values and  $d = g - x_n$  (nonzero on the essential degrees of freedom only),

$$J(x_n) s = R(x_n) + J(x_n) d, \quad x^{(0)} = x_n + d - s, \quad (54)$$

where  $R$  is evaluated with the loads at the new pseudo-time and  $s$  vanishes on the essential rows. Because the assembled Jacobian has its essential columns eliminated,  $J(x_n)d$  is formed by a forward difference of the residual,  $[R(x_n + \epsilon d) - R(x_n)]/\epsilon$ . This is the start that codes imposing the boundary condition on the Newton update obtain implicitly. The converged states are unchanged; Newton

typically needs half the iterations and no damped steps. The predictor matters most for nearly incompressible materials and is needed in practice for the logarithmic volumetric law, whose pressure  $\ln J/J$  is not monotone beyond  $J = e$  and whose stiffness  $u''$  varies by two orders of magnitude over the range of  $J$  of a lagging-interior iterate. It is a no-op for traction-driven increments, and it falls back to the default start if the predicted state has a non-finite residual.

## 12.2 Damped Newton

For  $R(x) = 0$  with Jacobian  $J(x) = \partial R / \partial x$  (`solvers/newton.hpp`):

1. evaluate  $R_0 = R(x_0)$ ; converged if  $\|R\| \leq \text{atol}$  or  $\|R\| \leq \text{rtol} \|R_0\|$ , where  $\|\cdot\|$  is the global  $\ell_2$  norm over the communicator and  $R_0$  is the residual at the start of the increment (after the Dirichlet data are applied);
2. solve  $J\delta = R$  with the linear solver; a Krylov solve that hits its iteration cap is reported and the inexact step is used;
3. Armijo backtracking: with  $\alpha = 1$ , accept  $x - \alpha\delta$  if  $\|R(x - \alpha\delta)\| \leq (1 - c\alpha)\|R(x)\|$  and the norm is finite, otherwise halve  $\alpha$ , at most `max_halvings` times ( $c = 10^{-4}$  by default). A non-finite trial residual (an inverted element, a Gent state beyond locking) is simply rejected;
4. stop with failure when the line search cannot reduce the residual or after `max_it` iterations; a problem that declares itself linear is instead accepted at the round-off floor of its residual (Section 7.5).

The report carries the iteration history and the failure reason, which the stepper uses to decide on bisection.

## 12.3 Linear solver of the displacement formulation

The Jacobian is a `HypreParMatrix`; the solver (`solvers/linear_solver.hpp`) is GMRES (default, restart `krylov_dim`) or CG, preconditioned by BoomerAMG with either the *elasticity* options (rigid-body-mode interpolation, needs the finite element space) or the plain *systems* options (unknown-based coarsening for  $d$  components). MFEM's Krylov tolerances are relative to the left-preconditioned residual  $\|M^{-1}r\|$ , which with AMG behaves like an error norm, so the true residual reduction is typically 10 to 100 times weaker than `rtol`. The elasticity options converge quickly in two dimensions but stall on slender three-dimensional  $k = 2$  meshes; they are given at most 100 iterations, after which the solver rebuilds the preconditioner with the systems options, repeats the solve, and keeps the systems options for the rest of the run. The AMG hierarchy is rebuilt for every Newton Jacobian, except for a linear problem, whose one Jacobian and hierarchy serve the whole load path (Section 7.5).

## 12.4 Saddle-point solver of the mixed formulation

Each Newton system has the structure (36),  $\mathcal{J} = \begin{bmatrix} \mathbf{K} & \mathbf{B} \\ \mathbf{B}_{pu}^T & -\mathbf{C} \end{bmatrix}$  with  $\mathbf{C} = \mathbf{M}_p/\kappa$  (zero when incompressible) and  $\mathbf{B}_{pu} = \mathbf{B}^T$  for the quadratic volumetric law only. When the pressure is comparable to the shear modulus the raw  $\mathbf{K}$  carries an indefinite  $p$ -weighted geometric stiffness, which AMG handles poorly. The solver (`solvers/saddle_point_solver.hpp`) therefore first rewrites the system in the algebraically equivalent *augmented Lagrangian* form: with  $\gamma = \text{augmentation} \times \mu$  and  $\mathbf{W} = \text{diag}(\mathbf{M}_p)^{-1}$ , adding  $\gamma\mathbf{B}\mathbf{W}$  times the second block row to the first gives

$$\begin{bmatrix} \mathbf{K} + \gamma\mathbf{B}\mathbf{W}\mathbf{B}_{pu} & \mathbf{B} - \gamma\mathbf{B}\mathbf{W}\mathbf{C} \\ \mathbf{B}_{pu} & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u} \\ \Delta p \end{bmatrix} = \begin{bmatrix} b_u + \gamma\mathbf{B}\mathbf{W}b_p \\ b_p \end{bmatrix}, \quad (55)$$

whose displacement block stays well conditioned (the augmentation acts as a penalty of strength  $\gamma$  on the discrete constraint). This is solved by flexible GMRES with the block upper-triangular preconditioner

$$\Delta p = -\tilde{\mathbf{S}}^{-1} r_p, \quad \Delta \mathbf{u} = \tilde{\mathbf{K}}^{-1} (r_u - \tilde{\mathbf{B}} \Delta p), \quad (56)$$

where  $\tilde{\mathbf{K}}^{-1}$  is an inner GMRES + BoomerAMG (systems options) solve of the augmented displacement block to `inner_rtol`, and  $\tilde{\mathbf{S}}$  approximates the Schur complement by a scaled pressure mass matrix,

$$\tilde{\mathbf{S}} = \frac{1 + \mu/\kappa}{\mu + \gamma} \mathbf{M}_p, \quad (57)$$

applied by CG with Jacobi smoothing. The scale interpolates between the classical estimates: without augmentation it is  $\mathbf{M}_p(1/\mu + 1/\kappa)$ , the sum of the incompressible Stokes-like estimate  $\mathbf{B}^T \mathbf{K}^{-1} \mathbf{B} \approx \mathbf{M}_p/\mu$  and the penalty term  $\mathbf{M}_p/\kappa$ ; in the incompressible limit it is  $\mathbf{M}_p/(\mu + \gamma)$ , the augmented-Lagrangian estimate. On the Cook and manufactured-solution problems this takes about 20 outer iterations independent of the mesh. `augmentation: 0` disables the transformation. The shear modulus that scales  $\gamma$  and  $\tilde{\mathbf{S}}$  is the base material's small-strain value (Section 4.6). The row operation multiplies by  $\mathbf{B}\mathbf{W}$ , the (1, 2) block itself, not by the transpose of the (2, 1) block; the two coincide only for the quadratic law, and the distinction is what keeps Newton quadratic with the other laws.

**Dynamic analysis.** The displacement block is then  $\mathbf{K} + c_M \mathbf{M}$ , and for small time steps the mass dominates it. The Schur complement of a mass-dominated block is a discrete pressure Laplacian,  $\mathbf{B}_{pu}(c_M \mathbf{M})^{-1} \mathbf{B}$ , which a scaled pressure mass does not represent: on the quarter annulus of the tube problem the outer iterations rise from 18 (quasi-static) to 43 at  $\Delta t = 0.002$  and, one refinement on, to 81. Following Cahouet and Chabard [15] the two limits are added,

$$\tilde{\mathbf{S}}^{-1} = \left[ \frac{1 + \mu/\kappa}{\mu + \gamma} \mathbf{M}_p \right]^{-1} + \left[ \frac{1}{c_M} \mathbf{B}_{pu} \mathbf{D}^{-1} \mathbf{B} + \mathbf{C} \right]^{-1}, \quad (58)$$

with  $\mathbf{D}$  the lumped mass (the diagonal of  $\mathbf{M}$  scaled to the total mass, positive on every element type and order, which row sums are not) and the second inverse by CG with BoomerAMG. The second matrix is formed from the blocks of the Jacobian at hand, so it carries their boundary conditions (prescribed displacements: Neumann; free surfaces: Dirichlet) without a pressure Laplacian being assembled. The count becomes 6 to 16 over three decades of  $\Delta t$ , independent of the mesh. The term exists only when the dynamic problem hands the solver  $c_M$ .

## 13 Post-processing

Two kinds of output fields exist (`physics/quadrature_fields.hpp`). The nodal unknowns, `displacement` and (mixed) `pressure`, are the  $H^1$  fields the solver computes and are written as they are. The quadrature quantities are computed once per load step at the quadrature points of the kernels' rule into a `QuadratureFunction`, the source of truth, from the state  $(\mathbf{F}, \mathbf{P}, \Psi)$  at each point and the measures of the material's kinematics formed from it (finite strain below; for a small-strain material  $\boldsymbol{\sigma} = \mathbf{P}$ , the volume ratio is  $1 + \text{tr } \boldsymbol{\epsilon}$  and the strain is  $\boldsymbol{\epsilon}$ , Section 7):

| Name                              | Components                     | Definition                                                                                                                                    |
|-----------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>cauchy_stress</code>        | 6 ( $xx\ yy\ zz\ xy\ yz\ xz$ ) | $\boldsymbol{\sigma} = J^{-1} \mathbf{P} \mathbf{F}^T$ , symmetrised                                                                          |
| <code>pk1_stress</code>           | 9 (row-major)                  | $\mathbf{P}$                                                                                                                                  |
| <code>deformation_gradient</code> | 9 (row-major)                  | $\mathbf{F}$ (completed $\mathbf{F}_c$ under plane stress)                                                                                    |
| <code>strain</code>               | 6 ( $xx\ yy\ zz\ xy\ yz\ xz$ ) | Green–Lagrange $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$                                                               |
| <code>jacobian</code>             | 1                              | $J = \det \mathbf{F}$                                                                                                                         |
| <code>vonmises</code>             | 1                              | $\sigma_{vm} = \sqrt{\frac{3}{2} \boldsymbol{\sigma}' : \boldsymbol{\sigma}'}$ , $\boldsymbol{\sigma}' = \text{dev sym } \boldsymbol{\sigma}$ |
| <code>energy_density</code>       | 1                              | $\Psi(\mathbf{F})$ ; mixed: $\Psi_{\text{iso}} + p(J - 1) - \kappa u^*(p/\kappa)$                                                             |
| <code>thickness_stretch</code>    | 1                              | $F_{33} = \lambda_3$ (plane stress only)                                                                                                      |

In two dimensions  $\mathbf{F}$  carries  $F_{33}$ , so the out-of-plane terms are included. Derived scalars are formed at the quadrature points and then presented, never computed from a presented field. Each quantity is presented where `output.quadrature_at` asks:

**nodes** a continuous  $H^1$  field of the mesh order. **nodal\_projection:** **averaged** projects the data onto each element’s own polynomial space (element mass matrix  $\mathbf{M}_e a = \int N f$ ) and takes the arithmetic mean of the element values at shared nodes (the classical extrapolate-and-average, communicated across ranks); **projected** solves the global  $L^2$  projection with the consistent mass matrix (CG to  $10^{-14}$ ), which is optimal in the least-squares sense but can overshoot near stress concentrations.

**elements** the quadrature-weighted volume average per element, an order-0 field named `<name>_elem`.

**quadrature\_points** the raw values, written as a point cloud `<name>_qp` (one VTU per rank, listed in the same PVD as the mesh).

For a homogeneous state all presentations agree to machine precision; on a real problem their differences are discretisation error. Probes evaluate every registered grid function (nodal unknowns, `<name>` and `<name>_elem`) at physical points, averaging when several ranks find the point. The internal energy  $\int \Psi dV$  (or the mixed functional) is printed at the end; it is computed by a domain-only form, so follower pressure work is not included (that load is not conservative). For the mixed formulation the integrand is  $\Psi_{\text{iso}} + p(J - 1) - \kappa u^*(p/\kappa)$ , where  $u^*(\pi) = \sup_J [\pi(J - 1) - u(J)]$  is the Legendre transform of the normalised volumetric law ( $\pi^2/2$  for the quadratic law, so that the integrand is the perturbed Lagrangian (35); closed forms for the Simo–Taylor and  $J \ln J$  laws, a scalar Newton solve for the logarithmic one). The stored energy  $\Psi_{\text{iso}} + U(J)$  is deliberately not used: the discrete displacement satisfies the constraint only weakly, and  $\kappa$  times the square of its pointwise violation would dominate the value for nearly incompressible materials. The last term is absent when  $\kappa = \infty$ , where the integrand is the Lagrangian and equals  $\Psi_{\text{iso}}$  on the constraint.

In a dynamic analysis the nodal fields **velocity** and **acceleration** of the time integrator join the unknowns (written and probed like the displacement), **output.energy** prints after every step the kinetic and the internal energy, the external work and their balance (Section 8), and **output.every** writes every  $n$ -th step to ParaView (and always the last); the probe, reaction and energy lines stay per step, which is how time histories are read from a log.

## 14 Mesh input

Meshes come from files (Gmsh 2.2 or 4.x, or any format MFEM reads). Every physical group becomes an element or boundary attribute numbered by its tag, and the group names are kept as attribute sets, so boundary conditions and material regions may name groups instead of numbers; both are checked against the mesh and the error lists what the mesh provides. The loader constructs the mesh in place because MFEM’s move operations drop the attribute sets. Options: **serial\_**

`refine` and `parallel_refine` uniform refinements (a second-order Gmsh mesh refines along the curved geometry, independently of the finite element order), and `perturb`, a deterministic pseudo-random jitter of the interior vertices by at most the given fraction of the smallest element size in each coordinate (straight-sided meshes only, applied before refinement so that all levels share one distortion pattern; inverted elements are refused). The Cartesian box and the bilinear corner map remain available programmatically for the tests.

## 15 Framework architecture and state of the seam

The architecture document defines a *flux/source seam*: a physics declares a flux  $\mathcal{F}(u, \nabla u)$ , a source  $\mathcal{S}(u)$  and a numerical flux  $\hat{\mathcal{F}}(u^-, u^+, \mathbf{n})$  per quadrature point, and the framework turns one definition into CG volume, DG face or HDG trace operators, with partial or full assembly and device execution. Solid mechanics fits this shape with  $\mathcal{F} = \mathbf{P}(\mathbf{F})$  in total Lagrangian form. The present code implements the seam as the minimum the solid needs; the following is still solid-specific and would generalise with a second physics:

- The displacement integrator hard-codes the flux  $\mathbf{P}(\mathbf{F})$  contracted with  $\text{Grad } \mathbf{w}$ , a vector  $H^1$  unknown and the contraction  $\text{Grad } \mathbf{w} : \mathbb{A} : \text{Grad } \Delta \mathbf{u}$ . The quadrature-point bodies are free functions on plain tensors and can be lifted into a general  $\mathcal{F}(u, \nabla u)$ ,  $\mathcal{S}(u)$  contract; the element loops would become the framework's CG volume kernel.
- The mixed kernel is a second block integrator with its own loops; a multi-field CG kernel over a list of unknowns with a block flux/source contract would absorb both. The saddle-point solver is specific to the  $2 \times 2$   $u$ - $p$  structure and the pressure-mass Schur approximation.
- Sources are not part of the integrator: the body force is a dead load on the linear-form side. A reaction or heat source needs a domain source term inside the nonlinear form with its own tangent.
- Boundary terms are stock linear-form integrators plus one hand-written face integrator (the follower pressure). Robin and flux conditions and DG/HDG numerical fluxes have no home yet.
- Dirichlet conditions prescribe components of the vector unknown on an attribute; scalar unknowns are not expressible in the input's `bcs` block.
- The physics owns its space, essential dofs and loads. A coupling layer needs these behind the common interface (`QuasiStaticProblem`: residual, Jacobian, pseudo-time, Dirichlet application) plus field exchange by name through the field registry.
- Time integration is a decorator over that interface, not a term of the kernels: with the unknown of the new time level as the variable, a second-order system adds  $c_M \mathbf{M}$  and a history vector to the static residual (Section 8). A first-order physics (heat, species) would add  $c \mathbf{C}$  and its history in the same way, with the first-order generalized- $\alpha$  parameters and the same  $\rho_\infty$ , which is what a monolithic coupling of the two needs. The mass matrix is assembled by a stock MFEM integrator here because it is constant; a capacity that depends on the state would have to enter the nonlinear form as a source kernel.
- Materials are stateless, one model per problem with region-wise parameters; internal variables, temperature dependence and hand-coded tangents are absent by design. The kinematics (finite or small strain) is a trait of the material, not of the kernel: the flux contract is shared, and what depends on the kinematics outside the flux goes through `materials/kinematics.hpp`. An eigenstrain (thermal, shrinkage) would enter where the small-strain material forms  $\boldsymbol{\varepsilon}$ , once materials can read a field.
- All kernels are CPU host code written without allocation or virtual calls in the quadrature

loops, so device annotations and `mfem::forall` can be added without restructuring; partial assembly is not implemented.

## A Symbols and input keys

| Symbol                                              | Meaning                                    | Input key                                                                        |
|-----------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------|
| $k$                                                 | polynomial order of the displacement space | <code>mesh.order</code>                                                          |
| $E, \nu$                                            | Young's modulus, Poisson ratio             | <code>material.E</code> , <code>material.nu</code>                               |
| $\mu, \kappa$                                       | shear and bulk modulus                     | <code>material.mu</code> , <code>material.kappa</code>                           |
| $u(J)$                                              | volumetric law $U = \kappa u$              | <code>material.volumetric</code>                                                 |
| $c_1, c_2$                                          | Mooney–Rivlin constants                    | <code>material.c1</code> , <code>material.c2</code>                              |
| $c_{10}, c_{20}, c_{30}$                            | Yeoh constants                             | <code>material.c10</code> , <code>c20</code> , <code>c30</code>                  |
| $J_m$                                               | Gent limit of $\bar{I}_1 - 3$              | <code>material.Jm</code>                                                         |
| $N$                                                 | Arruda–Boyce links per chain               | <code>material.N</code>                                                          |
| $\mu_r, \alpha_r$                                   | Ogden terms                                | <code>material.mu_r</code> , <code>material.alpha_r</code>                       |
| $\rho_R$                                            | reference density, also by region          | <code>material.rho0</code>                                                       |
| $t_{\text{final}}, \Delta t$                        | end time, time step                        | <code>dynamics.t_final</code> , <code>dt</code>   <code>steps</code>             |
| $\beta, \gamma; \alpha; \rho_\infty$                | time integration parameters                | <code>dynamics.scheme</code> and its keys                                        |
| $\mathbf{u}_0, \mathbf{v}_0$                        | initial displacement, velocity             | <code>dynamics.initial</code>                                                    |
| $\bar{\mathbf{u}}, \bar{\mathbf{T}}, p, \mathbf{b}$ | prescribed data                            | <code>bcs.dirichlet</code> , <code>bcs.traction</code> , <code>body_force</code> |
| $s_i(t)$                                            | load schedule                              | schedule of an entry                                                             |
| $t_1, \dots, t_n$                                   | load-step breakpoints                      | <code>solver.load_steps</code> or <code>solver.steps</code>                      |
| $x^{(0)}$                                           | initial guess of an increment              | <code>solver.predictor</code>                                                    |
| <code>rtol</code> , <code>atol</code> , $c$         | Newton tolerances, Armijo constant         | <code>solver.newton</code>                                                       |
| $\gamma/\mu$                                        | augmentation parameter                     | <code>solver.linear.augmentation</code>                                          |

## B Source files

| File                                       | Content                                                                                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| base/tensor.hpp                            | <code>tensor&lt;T, n...&gt;</code> of rank 1, 2, 4; products, contractions, transpose, trace, deviator, determinant and inverse up to $3 \times 3$          |
| base/dual.hpp                              | forward-mode dual number and elementary functions                                                                                                           |
| base/config.{hpp,cpp}                      | YAML schema, validation with key paths, schedules                                                                                                           |
| base/expression.{hpp,cpp}                  | parser and evaluator of $f(x, y, z, t)$                                                                                                                     |
| base/coefficients.hpp                      | expression and affine MFEM coefficients                                                                                                                     |
| base/mesh_input.{hpp,cpp}                  | mesh loading, attribute resolution, jitter, refinement                                                                                                      |
| base/fields.hpp, output, probes            | field registry, ParaView writer, point probes                                                                                                               |
| kernels/total_lagrangian.hpp               | quadrature-point functions and the displacement integrator                                                                                                  |
| kernels/mixed_total_lagrangian.hpp         | $u$ - $p$ quadrature-point functions and block integrator                                                                                                   |
| kernels/follower_pressure.hpp              | follower-pressure face integrators (both formulations)                                                                                                      |
| kernels/materials/*.hpp                    | material functors (hyperelastic, <code>linear_elastic.hpp</code> ), isochoric helpers, volumetric laws, spectral solver, plane-stress adapter, dual tangent |
| kernels/materials/kinematics.hpp           | finite or small strain as a trait of the material: Cauchy stress, volume ratio, strain                                                                      |
| kernels/materials/materials.{hpp,cpp}      | material variants, factory, moduli resolution, region tables                                                                                                |
| physics/loads.{hpp,cpp}                    | load set: Dirichlet entries, dead loads, follower scales, schedules, reactions                                                                              |
| physics/solid_mechanics_t1.{hpp,cpp}       | displacement formulation module                                                                                                                             |
| physics/mixed_solid_mechanics_t1.{hpp,cpp} | mixed formulation module                                                                                                                                    |
| physics/solid_problem.{hpp,cpp}            | common interface, factory by formulation, YAML load installer                                                                                               |
| physics/quadrature_fields.{hpp,cpp}        | quadrature quantities and their presentations                                                                                                               |
| physics/dynamic_solid_problem.{hpp,cpp}    | inertia as a decorator over a solid problem: mass matrix, step equation, initial state, energies, reactions with inertia                                    |
| solvers/time_integration.{hpp,cpp}         | parameters of the generalized- $\alpha$ family, spectral radius at infinity                                                                                 |
| solvers/quasi_static.{hpp,cpp}             | load stepping and time stepping with bisection                                                                                                              |
| solvers/newton.{hpp,cpp}                   | damped Newton with Armijo backtracking; acceptance of a linear problem at the floor of its residual                                                         |
| solvers/linear_solver.{hpp,cpp}            | GMRES/CG + BoomerAMG with the systems fallback                                                                                                              |
| solvers/saddle_point_solver.{hpp,cpp}      | augmented-Lagrangian FGMRES for the $u$ - $p$ Jacobian; inertial part of the Schur complement approximation                                                 |

## References

- [1] J. Bonet, R. D. Wood, *Nonlinear Continuum Mechanics for Finite Element Analysis*, 2nd ed., Cambridge University Press, 2008.
- [2] G. A. Holzapfel, *Nonlinear Solid Mechanics: A Continuum Approach for Engineering*, Wiley, 2000.
- [3] R. W. Ogden, *Non-Linear Elastic Deformations*, Ellis Horwood, 1984 (Dover, 1997).
- [4] J. C. Simo, R. L. Taylor, Quasi-incompressible finite elasticity in principal stretches: continuum basis and numerical algorithms, *Comput. Methods Appl. Mech. Engrg.* 85 (1991) 273–310.
- [5] T. J. Pence, K. Gou, On compressible versions of the incompressible neo-Hookean material, *Math. Mech. Solids* 20 (2015) 157–182.
- [6] S. Doll, K. Schweizerhof, On the development of volumetric strain energy functions, *J. Appl. Mech.* 67 (2000) 17–21.
- [7] E. M. Arruda, M. C. Boyce, A three-dimensional constitutive model for the large stretch behavior of rubber elastic materials, *J. Mech. Phys. Solids* 41 (1993) 389–412.
- [8] A. N. Gent, A new constitutive relation for rubber, *Rubber Chem. Technol.* 69 (1996) 59–61.
- [9] O. H. Yeoh, Some forms of the strain energy function for rubber, *Rubber Chem. Technol.* 66 (1993) 754–771.
- [10] L. R. Herrmann, Elasticity equations for incompressible and nearly incompressible materials by a variational theorem, *AIAA J.* 3 (1965) 1896–1900.
- [11] M. Benzi, M. A. Olshanskii, An augmented Lagrangian-based approach to the Oseen problem, *SIAM J. Sci. Comput.* 28 (2006) 2095–2113.
- [12] N. M. Newmark, A method of computation for structural dynamics, *J. Eng. Mech. Div. ASCE* 85 (1959) 67–94.
- [13] H. M. Hilber, T. J. R. Hughes, R. L. Taylor, Improved numerical dissipation for time integration algorithms in structural dynamics, *Earthquake Eng. Struct. Dyn.* 5 (1977) 283–292.
- [14] J. Chung, G. M. Hulbert, A time integration algorithm for structural dynamics with improved numerical dissipation: the generalized- $\alpha$  method, *J. Appl. Mech.* 60 (1993) 371–375.
- [15] J. Cahouet, J.-P. Chabard, Some fast 3D finite element solvers for the generalized Stokes problem, *Int. J. Numer. Methods Fluids* 8 (1988) 869–895.
- [16] R. Anderson et al., MFEM: A modular finite element methods library, *Comput. Math. Appl.* 81 (2021) 42–74.